



**Kushal Kafle**

**2133095**

## ***Smart Irrigation System***

**BSc (Hons) Computer Science and Software Engineering**

**Undergraduate Final Thesis Report**

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Department of Computer Science & Technology**

**University of Bedfordshire**

**Supervisor:**

**Mr. Krishna Aryal**

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Student's Surname  
**Kafle**

Student's Forename  
**Kushal**

Unit Leader's Name:  
**Ajay Sharma**

Supervisor:  
**Krishna Aryal**

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## Abstract

The research on Smart Irrigation System focuses on incorporating the technologies like IoT and AI for the development of smart and intelligent systems that can be implemented to enhance the agricultural practices and improve the irrigation systems, which in turn helps tackle the water scarcity issues without compromising the crops' water requirements. The research focuses on finding the right IoT devices, assembling them and then integrating them with the remote AI models as per the requirement of the project. Given that the traditional irrigational systems face various climatic challenges and are also inefficient while managing the water resources, this research aims to find the appropriate and innovative approach to improve and optimize the water consumption in agricultural fields by the help of precision irrigation systems based on AI and IoT. The research emphasis on the development of a smart and intelligent irrigation system which is scalable and can adapt to various agricultural settings and environmental conditions. Another main goal of the research is to find the right sensors and AI algorithms that are suitable for the project requirements. Additionally, to accurately estimate the water usage by the crops, it is necessary to find right mathematical approaches and formulas, to calculate and estimate the water flow from the system in given time period. The system interacts with various sensors to gather real time data on soil moisture, temperature and humidity and also keeps track of the crop age. The collected sensor data along with the crop age is then analyzed by the AI model which makes decision on whether to start or stop the irrigation. System also measures the amount of time taken by each irrigation cycles in order to track the water usage. The system then presents the water consumption statistics by various crops over their lifetime which helps farmers make estimates about the water requirement of each crop and plan accordingly. This helps to improve the crop production as they can select crops based on current water availability and also helps reduce water wastage.

Keywords: ESP32 based Irrigation, AI in irrigation, Smart Irrigation, Internet of Things in Agriculture, Predictive Irrigation System

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## Abbreviations

|         |                                            |
|---------|--------------------------------------------|
| IoT     | Internet of Things                         |
| AI      | Artificial Intelligence                    |
| ML      | Machine Learning                           |
| PLSR    | Partial Least Square Regression            |
| LoRaWAN | Long Range Wide Area Network               |
| RTD     | Resistance Temperature Detector            |
| LoRa    | Long Range                                 |
| UV      | Ultraviolet                                |
| IR      | Infra-Red                                  |
| CSIS    | Conventional-Scenario Intelligent-Scenario |
| SVM     | Support Vector Machine                     |
| KNN     | K-Nearest Neighbors                        |
| RMSE    | Root means square deviation                |
| MQTT    | Message Queuing Telemetry Transport        |
| LCD     | Liquid Crystal Display                     |
| LSTM    | Long Short-Term Memory                     |

# Chapter 1 : Introduction

## Background

Efficient irrigation in agriculture one of the pressing global challenges, further worsen by the climate change, rapid population growth and increasing food demand. The global population is expected to reach 9.7 billion by 2050 which requires a 70% increase in agricultural production to meet the global food demands (UN Report 2019). Agriculture is already accounted for about 70% of global freshwater consumption, making it crucial to adopt strategies and solutions that can optimize the water usage without compromising the productivity (Vallejo-Gómez et al. 2023). Traditional irrigation methods often overuse the water resource thereby increasing the water scarcity issues which threatens the global food security and sustainable agriculture.

The beginning of smart technologies like IoT and AI and their synergy offers innovative and promising solutions to these adverse challenges as they enable the precise and automated irrigation system that can adapt to the real time environment and crop specific conditions and settings. These technologies are found to optimize the water usage as well as enhance the crop production making them vital to address the water and food challenges in modern agriculture.

However, these technologies are a bit expensive and are not easily available for the farmers in the rural areas. Hence this thesis focuses on exploration of cost-effective, scalable and easily available smart irrigation solutions. The motivation lies in developing an easily adaptable and practical system that integrates AI and IoT to demonstrate their combined potential for sustainable irrigation. Confined within the resource constrained environment, the project aims to use simple, cheaper and effective components to provide actionable insights for local farmers through real time monitoring and control.

## Problem Statement

The rapid urbanization and the significantly increasing global population have increased the global food demand requiring the advancements in agricultural sectors to ensure the food security. The challenge is accompanied by the water scarcity, as agriculture alone accounts for 70% of global freshwater consumption (FAO, 2011). Additionally, the human land occupancies are also increasing for urbanization purposes which in turn reduces the arable land, further stressing the agricultural production (Doshi, Patel, and Bharti, 2019). Traditional irrigation systems cannot precisely address this issue often leading to under irrigation which reduces the productivity or over irrigation which wastes the water and also decreases soil fertility (Khanna and Kaur, 2019).

Technologies like the Internet of Things (IoT) have an immense scope to influence agriculture by enabling irrigation control in real-time and thereby make water usage much more efficient. Such systems are able to use water more efficiently, decrease human involvement, and raise agricultural production (Wolfert et al., 2017). Nevertheless, the financial constraints, farmers'

lack of technical knowledge, and complexities of setting up such projects in rural areas without adequate infrastructure are the factors that delay the implementation of such technologies (Bwambale, Abagale, Anornu, 2022).

## Proposed Solution

The basic approach is to explore the feasibility of a low cost smart irrigation system that uses a microcontroller, soil moisture, temperature and humidity sensors and a simple water pump. A simple AI model shall be used. It should be capable of taking the sensor data, crop type and crop age as the input and then predict whether to irrigate the field or not depending on the input data.

This approach draws the inspiration from reviewed studies that focuses on real time monitoring and automation. For instance, those irrigation systems that utilize the IoT with embedded devices demonstrated the reliability, potential and effectiveness of low cost solutions for remote data collection and control (Sitharthan et al. 2023; Kurundkar et al. 2023). Also, the integration of predictive AI models ensures that the field and crops are irrigated efficiently based on real time environmental data and forecasted whether conditions (Choudhary et al. 2019). This approach eliminates the need of advance and costly technologies, making it feasible within the limited resources.

By focusing on the simple AI and scalable IoT components, this approach achieves the balance between innovation, practicality and effectiveness. It contributes to the water conservation and sustainable farming and demonstrates the potential and applicability of smart agricultural technologies.

## Aims And Objectives

### Aims

The main aim of the research is to explore the application of IoT and AI to address the problems connected with the traditional irrigation methods. Research aims to support the development of more efficient, less expensive, and even sustainable irrigation systems, it aims to offer water- saving, crop production increasing, and soil-specific adaptive irrigation systems that can be eco- friendly, cost-efficient. In summary, the research aims to explore approaches to optimize the irrigation systems to tackle global water scarcity and to increase the agricultural productions in order to ensure food security.

### Objectives

- Research and implement one of the cost-effective methods for smart irrigation using AI and IoT that is feasible even in resource constrained environment.
- Build a fully functional precision irrigation system using esp32 integrated with real time clock and sensors to control the irrigation based on real time sensor inputs.
- Integrate a simple AI model to process and analyze the sensor data and crop age and accurately predict the irrigation needs.

- Measure the water consumption by each crop throughout its life span until harvesting.
- Implement user friendly dashboard to control and monitor the system.

## Feasibility Study

For the successful development and implementation of the Smart Irrigation System, factors like financial, technical and operational feasibility needs to be carefully considered. In context of this project, since the resource are limited to basic IoT components, it is important to evaluate the viability of achieving the expected goals of water conservation and improved irrigation efficiency.

### Technical Feasibility

The proposed system uses the existing and experimentally proven IoT technologies and sensor- based automation which ensures reliability as well as compatibility. Several studies have proven that the microcontroller-based systems are often efficient in data processing and wireless communication thus being suitable for resource constrained environments (Doshi, Patel, and Bharti, 2019). Also, by incorporating the temperature, humidity and soil moisture sensor for precise monitoring and decision making, the system meets the recommendations for data driven irrigation system (Khanna and Kaur, 2019).

### Financial Feasibility

One of the main difficulties of smart irrigation technologies is the high cost, which makes them unaffordable for small-scale farmers (Bwambale, Abagale, and Anornu, 2022). This project tries to solve this problem by using cheaper yet effective IoT components. Also, the use of open-source platforms like the Arduino IDE reduces the costs of development significantly. Earlier research reveals that the application of such cost effective IoT systems in agriculture farms can bring significant returns by saving water and increasing crop harvesting which encourages farmers to adopt such technologies (Wolfert et al., 2017).

### Operational Feasibility

The system design focuses on ease of use and simplicity aiming to make it accessible to the farmers with limited technical knowledge. However, the initial setup may still require some technical assistance. Features like automated irrigation based on real-time data along with the manual override options via a web app enable the user to have freedom of choice thereby increasing user acceptance. Additionally, this system is scalable which makes it possible to introduce it in various farm settings based on the crops types. Thus, it is possible to cover a wide range of different agricultural sectors. Nevertheless, problems such as rural areas having weak internet connectivity may cause real-time monitoring to be limited thus offline data storage and processing should be the ways out (Khanna and Kaur, 2019).

## Environmental Feasibility

Water shortages are still one of the most severe worldwide problems. This holds true especially in areas where agriculture uses majority of the available water resources (FAO, 2011). The suggested approach is a water sustainability program through the optimization of watering strategies synchronized with the soil and weather conditions. Besides, this also ensures better crop health and productivity, which benefits the environment promoting sustainable agriculture.

## Intellectual Challenge

The development of smart irrigation system using IoT and AI faces several intellectual challenges that needs attention and careful considerations to ensure its usability and effectiveness. It is indeed very challenging and complex task to integrate the modern technologies into the domain like agriculture which has been traditionally manual over a long time. Additionally, working withing the constraints of limited resources and everchanging environmental conditions further escalates the complexities.

## Data Acquisition and Interpretation

One of the main intellectual challenges is acquiring and processing environmental data like soil moisture, temperature, and humidity, which is the prerequisite for making precise irrigation decisions. The variations in these parameters among various crops, soil types, and weather conditions, on the one hand, system requires the development of adaptive algorithms that include capabilities of dealing with diverse scenarios, while on the other hand, makes it challenging to develop a comprehensive model (Khanna and Kaur, 2019). Also ensuring sensor accuracy by mitigating noise in the data acquisition add another layer of challenge and complexity of the project.

## AI model development and resource constraints

Design and implementation of an AI model efficient enough to accurately predict irrigation requirements of the crop and fields based on the limited parameters like temperature, humidity, soil moisture and crop age is yet another challenge, as other environmental factors like sunlight, weather conditions, soil fertilization needs etc may impact the irrigation needs and also heavily impact the productivity. Besides due to the lack of hardware with high computational power, AI model must be lightweight and efficient which requires innovative approach to minimize the computational needs while achieving the higher prediction accuracy (Liakos et al., 2018).

## Scalability and Adaptability

Creating a system that is easily adapted to different farm sizes, soil types, and crop requirements is another challenge, which requires innovation. The approach must be not only economical but also feasible in order for it to be a viable solution for small-scale farmers. The

system with the feature of manual control and automation via app gives more adaptability but increases complexities in system design (Tzounis et al., 2017).

### Environmental and Connectivity Challenge

The system relies on IoT technologies which makes it vulnerable to connectivity issues especially in rural areas where the internet connectivity is weak and slow. This requires designing a system that can operate offline and then sync data when the network connectivity is available (Obaideen et al., 2022). Also the system needs to consider the mechanism to handle unpredictable environmental conditions like unexpected rainfall, or other extreme weather conditions (Choudhary et al., 2019).

### Promoting User Adoption

Finally, the difficulty of integrating this method into farming activities faces a socio-technical challenge. Most farmers might not have the necessary skill set and thus, thinking about the design of a user-friendly interface and provision of adequate training becomes a priority to encourage adoption. Thus, balancing the simplicity, usability and functionality without hindering the system performance is the main part of the project's success (Liakos et al., 2018).

## Chapter 2 : Project Plan

### Gantt chart

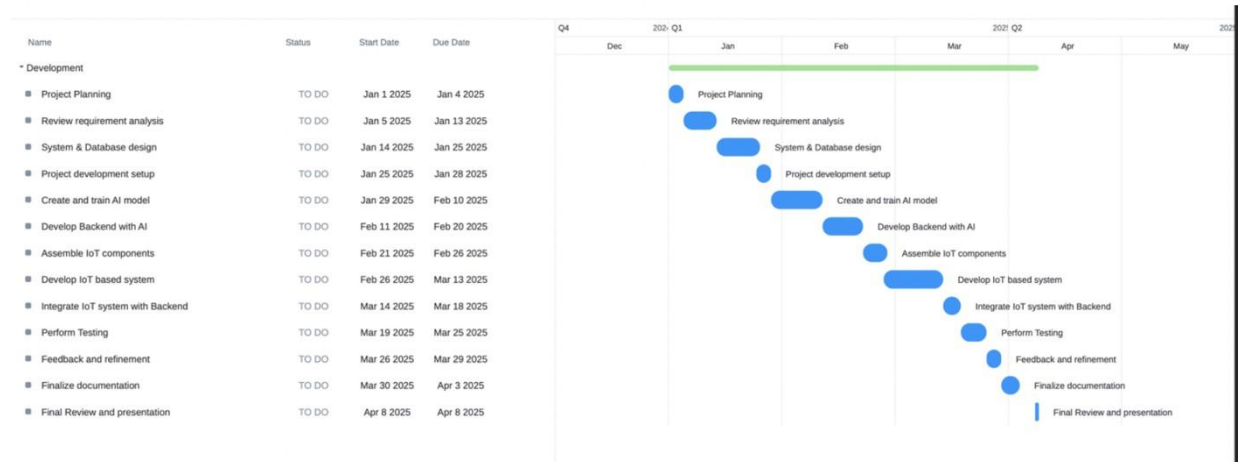


Figure 1:Initial Gantt chart

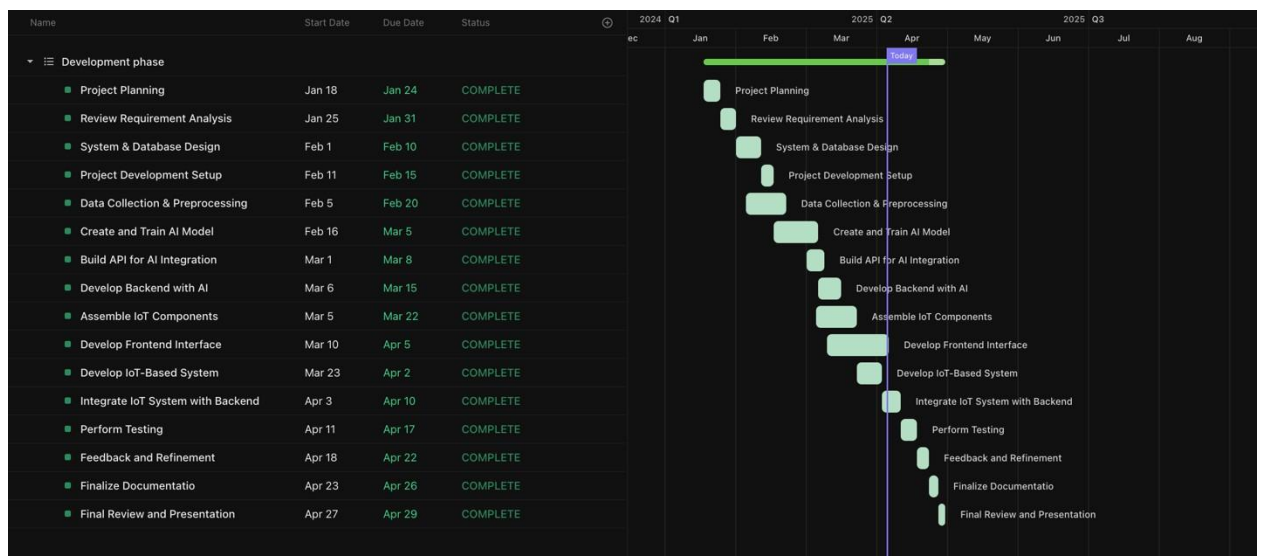


Figure 2: Revised Gantt chart

### Risk Analysis

The risk analysis is the process of identifying and evaluating the potential risk factors or challenges that can possibly affect the implementation of the smart irrigation system. The process tries to assess the likelihood and impact of these risks and consider the mitigation strategy for each of them as follow:

| Risk                        | Impact | Likelihood | Mitigation Strategy                                                                                                                      |
|-----------------------------|--------|------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Risk of Data Quality        | High   | High       | Perform data validation checks, by implement multiple sensors with options of data redundancy. Also Perform periodic sensor calibration. |
| Risk of Power Outage        | Medium | High       | Incorporate backup power sources for emergency, use uninterruptible power supplies or implement solar power.                             |
| Risk of sensor fault        | High   | Medium     | Use multiple sensors with data redundancy and cross validate the data. Also perform regular maintenance checks and sensor calibrations.  |
| Risk of unexpected rainfall | Medium | High       | Consider using weather forecasts data and irrigate accordingly.                                                                          |

*Table 1 :Risk Table*

## Work Break-down Diagram

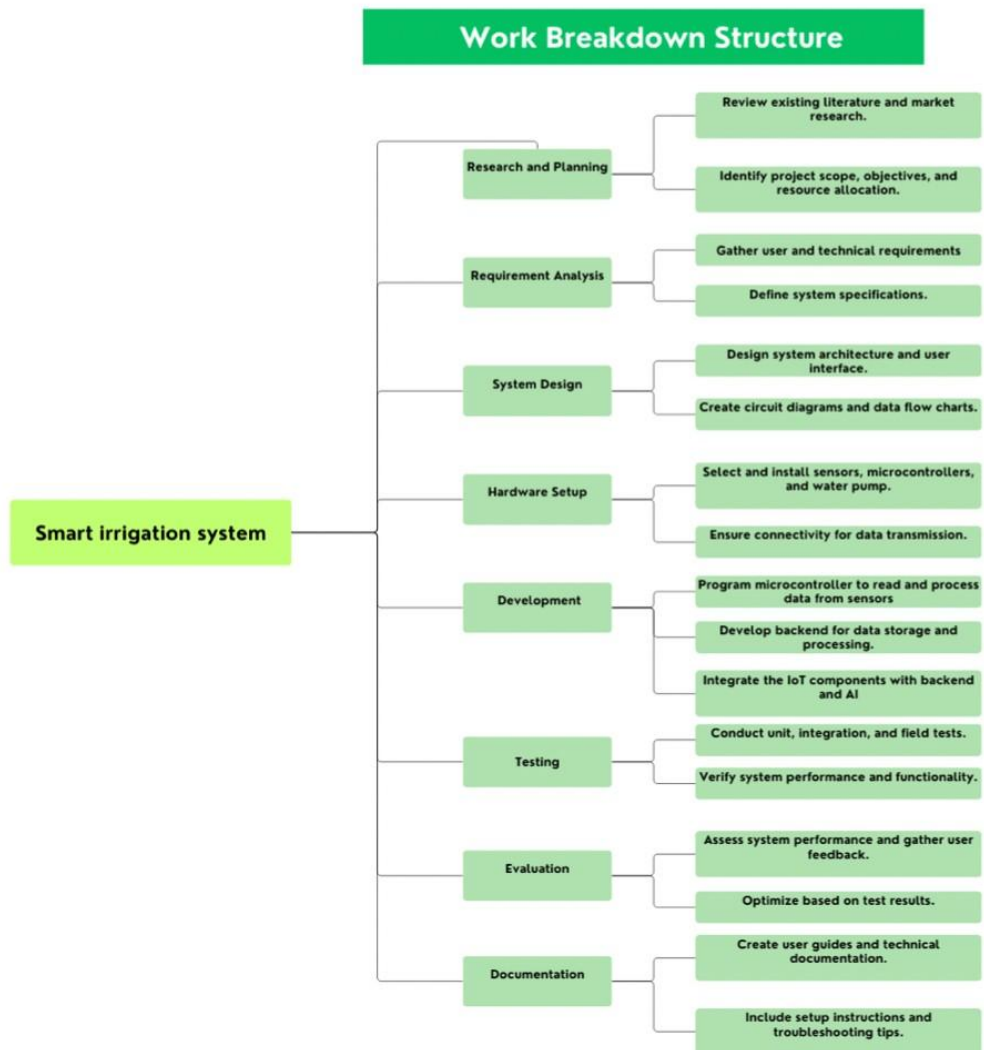


Figure 3: Work break down of the project

## Chapter 3 : Literature Review

Several researches and experiments have been conducted in order to test the potential of synergy of AI and IoT in addressing the agricultural water crisis and lower crops productivity. Most of the have demonstrated the promising results, some of which are reviewed below.

### Background

A literature review serves as a solid basis for any research project. It helps to gain the pivotal information on the thoughts, ideas and models currently in use, the gaps in the research, and the basis for the new study. In context of Smart Irrigation System by using AI and IoT is, the literature review is done for a number of reasons. It delves into the automation and shortcomings of the current intelligent irrigation systems, therefore, leading to the development of a sustainable and resource-saving solution through the comparison of the current technologies. The assessment of various methods, tools, and techniques of earlier studies helps to grasp the different schools and issues in the smart irrigation systems sector.

The literature review also helps benchmarking by comparing various existing technologies in the field. In context of smart irrigation, where resources are limited to components like Arduino, esp32 microcontrollers, basic sensors and simple AI models for binary decision making, it is highly significant to understand the trad-offs in existing systems. It makes certain that the proposed solution remains feasible while incorporating key advancements like AI model integration and calculating water usage statistics. Furthermore, the review process also justifies the need of this research by highlighting the limitations of existing systems such as high cost, reliance in commercial platforms and difficulty in implementing in rural areas.

Finally, the literature review is one of the vital means to fulfill gap between the research and the current trends while dealing with distinct problems and purposes of the newly proposed system. It assures the research is thoroughly based on the established knowledge of the domain and offers a roadmap for innovation in the context of precision irrigation.

### Research on Existing technologies

#### 1. A Novel Autonomous Irrigation System Using AI and 6G-IoT Networks

The study conducted by Sitharthan et al. (2023) introduces an autonomous irrigation system that combines 6G enables IoT and AI technologies. This solution addresses critical challenges in agriculture such as water wastage, inefficiency in traditional irrigation system and increasing demand for sustainable farming. By using AI prediction model integrated with microprocessor, the system utilizes the weather history and real time environmental data to make precision irrigation decisions.

By integrating the 6G network with IoT, this study has already contributed to revolutionize the agricultural sector, by providing faster, reliable and efficient solution for remote communication and monitor. A 6G IoT network is the next-generation wireless technology with ultra-high speeds, near zero latency and massive data transmission capacity.

The study emphasizes on a multi layered system, consisting of a physical, edge, cloud and network layers that are interconnected via 6G enabled IoT system. This makes the system efficient in data collection, processing and transmission. The data is collected from environment via different sensors like temperature and soil moisture and then transmitted to the cloud server for processing. The AI model then predicts the soil moisture requirement based on the given data and then adjusts the irrigation schedule accordingly thereby optimizing the water usage.

The experiment and evaluation were conducted in controlled environment. The system showed high performance achieving 86.34% accuracy, 89.28% sensitivity, and 91% precision (Sitharthan et al. 2023). These result shows the potential of integrating AI and smart IoT technologies in transforming traditional farming into sustainable agricultural practices. Remote monitoring feature via web-based interface also highlights the user centric design, making it easier for farmers to manage and monitor irrigation process from distant location.

Limitations:

The framework proposed by Sitharthan et al. (2023) provides a promising and comprehensive foundation for future advancement in smart farming. While the study offers a strong foundation for smart irrigation, its reliance on 6G network may compromise its immediate feasibility in rural regions which lack the proper internet connections. Besides, 6G networks and its equipments are often expensive which makes it unaffordable for farmers with financial constraints.

## 2. Autonomous Crop Irrigation System using AI

The study conducted by Choudhary et al. (2019) presents another autonomous irrigation system that combines AI and IoT to address the inefficiencies in traditional irrigation methods. The study focuses on optimizing the agricultural water usage and enhancing the crop production by leveraging ML models and real time environment data. As the agriculture is a critical sector in India characterized by inefficient, unorganized and haphazard irrigation techniques, the study proposed this system as a solution which holds significant potential for improving water conservation along with the productivity.

Autonomous irrigation system developed by Choudhary et al. (2019) uses NodeMCU, soil moisture sensor and Raspberry pi 3B+ as a central controller. It monitors and collects the soil, and weather conditions and then processes the data at the interval of 30 minutes. The system uses Partial Least Square Regression (PLSR) to determine irrigation needs and Random Forest Regressor for rainfall prediction which enhances the system's accuracy and adaptability to various environmental conditions and ensures the accurate prediction of soil moisture and water needs.

The irrigation requirements accuracy is further enhanced by using the historical rainfall data and real time sensor inputs. Furthermore, the system periodically updates its dataset in order to improve its predictive accuracy and performance, making the system self-learning that can adapt to local climatic variations over time. The system follows the distributed architecture in

which each node functions independently, hence the system keeps running even if one node fails. The system is powered by the solar panel which uses sun's energy, making the system Eco-friendly. Such features of the system make it properly align with the goals of sustainable and Eco-friendly agricultural practices.

Limitations:

Despite its core strength, the study still has certain limitations. It depends on external sources like government portals for the high-quality rainfall data, this might restrict its real time applicability in rural regions where such high quality reliable data is unavailable. Besides the absence of long- range communication systems, restricts its scalability in large agricultural sectors. Also, the system relies on Raspberry pi as central controller which may introduce bottlenecks while processing those data and managing multiple nodes simultaneously.

### 3. IoT based Precision Irrigation System for Tomato Production

The study presented by Seyar and Ahamed (2023), explores the use of IoT in precision irrigation systems specifically tailored for the tomato farming. It experiments on two methods of irrigation i.e. surface and sub surface drip irrigation in both indoor and outdoor environment with the focus on optimizing the water usage and enhancing tomato production.

The study emphasis on the importance of precision irrigation specially for water sensitive crops like tomatoes. By using the IoT technologies and low-cost soil moisture sensor, the researchers developed an irrigation system that automates the water delivery by monitoring the real time soil moisture, temperature and humidity. The study identifies the threshold of 12% soil moisture as ideal for healthy tomato growth on indoor germination. When the system was transitioned for the outdoor environment, they employed a LoRaWAN based system for remote control and monitor of irrigation system and then evaluate the result of both the irrigation methods based on crop performance. Subsurface drip irrigation was found to be more water efficient using 10% less than its counterpart i.e. surface irrigation. And also, subsurface irrigation produced higher crops i.e. 1243 gram/plant compared to 1061 gram/plant by the surface irrigation. The system makes the proper use of cloud platforms like ThingSpeak for real time data visualization and control. The research seems to be successful in providing a comprehensive evaluation of the system's impact on tomato growth and production by incorporating the agronomic parameters like root length, germination rate and plant yields.

Limitations:

As the findings of the research seems promising, there are still some limitations that needs to be put into perspective. The research only focused on small scale indoor and outdoor experiments that too with controlled conditions which may not reflect the large-scale agricultural fields complexities. Also, the subsurface irrigation faces the problem like clogging, and the heterogeneity is absence in field soil which makes the generalization of the results from the experiment questionable.

#### 4. Smart Irrigation System Based on IoT and Machine Learning

Tace et al. (2022) proposed an intelligent irrigation system utilizing IoT and Machine Learning, thus dealing with the problems of water scarcity and optimizing irrigation in agriculture. The system combines sensors (soil moisture, temperature, and rain) to develop a data collection mechanism through Node-RED and MongoDB platforms. The list of algorithms includes not only K-Nearest Neighbors (KNN) but also Logistic Regression, Neural Networks, Support Vector Machine (SVM) and Naïve Bayes, which were applied for data analysis. The method showed the best results with the KNN model which recorded an accuracy score of 98.3% and an RMSE of 0.12.

One of the major research contributions is that Tace et al. (2022) developed an intelligent irrigation system integrating AI and IoT in the field of agriculture and also incorporated remote monitoring of the system via web-based application. The experiment also shows that the higher accuracy can be obtained by using KNN algorithm for machine learning which outperforms other models. The system can conserve the resources through precise water management strategies.

##### Limitations:

The system consists of some limitations. The testing and evaluation of system was conducted in controlled environments which may not accurately represent the real-world challenges that occur in diverse agricultural settings. This makes the reliability and applicability of the system questionable when applied to large scale real world agricultural environments. Additionally, the cost effectiveness of the system is not mentioned clearly which acts as a barrier for adoption in small scale agriculture and farmers with financial constraints.

#### 5. IoT-Based Smart Irrigation System Using AI

Nanthakumar et al. (2024) came up with a smart irrigation system that uses IoT and AI technology to achieve precise water management. The system makes use of soil moisture sensors, relays, water pumps, and other equipment to monitor the environmental conditions and dynamically optimize water use most efficiently. The real time updates and alerts are provided to the user via user friendly user interface consisting of an LCD and mobile application which enhances the usability of the system.

The study proposed a comprehensive IoT-based model for precision agriculture that emphasis on resource preservation and system efficiency. The AI was empowered with soil moisture sensors to monitor the real-time data and analyze it to predict appropriate irrigation schedule which prevents the waste of water. The hardware design of the system was consisting of the IoT devices like Arduino Uno, ESP8266, Wi-Fi modules, and the relay modules to seamlessly control the water pumps. The integrated buzzer and LCD provide the feedback to the user about the system status and alerts. A mobile app interface was developed to remotely track the irrigation system, which enabled user to monitor soil moisture levels, pump activity, and irrigation history. The AI algorithms were trained to study the environmental factors like soil

conditions, weather patterns and crop specific requirements then dynamically adjust the irrigation frequency as per the requirement ensuring an adaptable and scalable system.

Limitations:

There are certain limitations in the system like the dependency on the stable internet for continuous data transfer and decision making may limit its applicability in rural areas with unstable internet connections. Also, the durability of the system is not tested properly, the long-term exposure to dust, heat and moisture in harsh agricultural environment may damage the system or at least affect its performance.

## 6. Intelligent IoT Sensor Coupled Precision Irrigation Model for Agriculture

Lakshmi et al. (2023) proposed a smart irrigation system using IoT and machine learning technologies for controlling water resources in agriculture. The system relies on soil moisture sensors and LoRa based wireless communication for real-time measurement, as well as an LSTM neural network to predict the irrigation requirements. Utilizing the environmental and soil data for the irrigation decision-making process, the system successfully managed to reduce water consumption without compromising the crop water requirement and also maintaining the crop health.

The research delivered an advance IoT based irrigation system that is scalable and resource efficient. Soil moisture sensors provide the real-time soil data, whereas LoRa communication ensures long-range data transmission with minimal power use. The system was integrated with the LSTM neural network which allowed it to accurately predict the irrigation requirements based on environmental parameters like soil moisture, air humidity, temperature, and the amount of rainfall, thus ensuring precise water management. The model achieved 46% reduction in water use, as confirmed by the field tests among crops like tomatoes and eggplants. Also, the system consists of a user-friendly graphical user interface for the remote monitoring via Blynk application which provides the farmers with the insights of the system like irrigation schedules, live system status and soil conditions.

Limitations:

There are certain limitations in the system that needs to be addressed. The tests were conducted in small plots and system's scalability to large scale farms with diverse agricultural settings are not evaluated properly. Also, the cost effectiveness of the system for small scale farmers is questionable. The system uses advance IoT technologies like LoRa modules and also the LSTM networks are computationally intensive which may not be feasible in resource constrained environments.

## 7. AIDSII: An AI-Based Digital System for Intelligent Irrigation

The paper by Raouhi et al. (2023) introduces a value-adding technology to agricultural irrigation through AI and the Internet of Things. The study aims to alleviate water scarcity and enhance crop productivity in the first place through automatic irrigation scheduling. Researchers decrease irrigation water use to sustainable levels by means of predictive AI with

real-time data collection using IoT devices, leading to sustainable and more efficient agricultural practices on farms.

The system uses a CNN-LSTM model for processing environmental data such as soil moisture, temperature, and humidity for predicting optimal irrigation schedules. The use of three separate layers, namely, application, intelligence, and hardware, makes the software exceptionally good at incorporating AI and IoT components. IoT hardware consists of soil moisture sensors and temperature probes, which provide data for the prediction model. The results are presented in a user-friendly way via mobile and web-based interfaces which can be accessed remotely. Controlled tests successfully showcased the system's efficiency, reducing water wastage and boosting crop yields by improving the precision of irrigation.

Limitations:

Nevertheless, the system's dependence on the credible Internet of Things (IoT) sensor data necessarily becomes vulnerable to hardware reliability issues; faulty sensors will stimulate the incorrect prediction. Besides, the technical difficulty probably will discourage adoption among small farmers who have no enough knowledge. Future research has to consider these difficulties by creating reliable methodologies to process noisy or missing data and thus furthering the system to add more functionality such as pest prediction for full-scale agricultural management.

## 8. IoT-Based Smart Irrigation System Using Artificial Intelligence

Rahul et al. (2022) suggested a smart irrigation system that integrates the Internet of Things (IoT) and Artificial Intelligence (AI) as an effective solution for handling water conservation problems in agriculture. The methods mentioned in the study involve the use of soil moisture sensors, prediction algorithms, and IoT communication devices to measure the level of environmental conditions and consequently do the irrigation automatically.

The collected information presents the framework of IoT and AI in precision irrigation, especially automation and resource optimization. Soil moisture sensors and the DHT11 temperature sensor was utilized together with an Arduino microcontroller and ESP8266 WiFi module to allow the real-time data collection and analysis. The system is granted power of intelligent automated scheduling of irrigations via weather forecasts and knowledge based soil conditions, by the help of precise and efficient prediction algorithms. Energy constraints in remote areas were solved through the use of solar power, which in turn, made the system more effective and sustainable. The method was evaluated in a closed control experiment and detected an accuracy of 80%. Besides, the study also highlighted the practical uses of IoT for reducing human involvement, as well as, enhancing crop health and supporting sustainable agricultural practices.

Limitations:

The research and the project presented by Rahul et al. (2022) exhibits several limitations that merit study and revision in the future. The first concern is that while the system is wisely

connected to the internet relying on Wi-Fi ESP8266 module, some areas with poor connectivity or constrained network infrastructure might limit the applicability of the system. On one hand, solar energy provides better electric efficiency; on the other hand, the initial cost for the solar system parts and the Internet of Things (IoT) devices may hinder the adoption by smaller-scale farmers or those with limited resources. Thirdly, the testing and evaluation was done in a controlled environment that might not be able to capture the full dynamics of the diverse agricultural world, like different soil types, weather conditions, or pest infestations. Moreover, the prediction algorithms produced only 80% accuracy which can be taken as an opportunity for improvement to get higher reliability in real-world circumstances.

### 9. Solar-Powered Smart Irrigation System Using Machine Learning & IoT

Ramprasad et al. (2024) recommended a solar-based smart irrigation system that utilizes IoT and machine learning to maximize water efficiency and to increase crop production. The system uses solar panels, IoT devices, sensors, and Random Forest algorithms to automatically carry out irrigations according to the weather condition. By means of real-time data from the sensors, it generates the most appropriate watering time and sends notification messages to users. The system reduces water usage and human involvement in the farm.

The study presented an innovative IoT-based irrigation system powered by renewable solar energy, ensuring sustainability and uninterrupted operation. Key components include photovoltaic panels for energy generation, DHT11 sensors for temperature and humidity monitoring, and ADS1115 analog-to-digital converters for interfacing with soil moisture and rainfall sensors. The Raspberry Pi microcontroller serves as the central unit, collecting sensor data and processing it using machine learning algorithms. The Random Forest algorithm analyzes multiple environmental factors, including soil moisture, temperature, and light intensity, to predict optimal irrigation schedules. Push bullet notifications provide real-time alerts to farmers about environmental conditions, enabling remote monitoring and management. The system demonstrated effectiveness in water conservation and adaptability to varying agricultural conditions.

#### Limitations

The system faces a few challenges that reduce its applicability as well as its potential to scale to different agricultural sectors. Although the Random Forest algorithm is one of the most precise techniques, it requires a large enough dataset for training, but in areas with little historical data or that suddenly experience environmental changes, the algorithm may decrease its performance. The increased cost for components like Raspberry Pi or sensors as well as IoT modules can be an impeding factor preventing small farmers lacking proper resources from investing in them. Lastly, the learning curve involved in the process of installing, configuring, and maintaining the system may be a source of the bottleneck for users with no prior knowledge of technologies like the Internet of Things (IoT) and machine learning.

## 10. IoT-Enabled Smart Irrigation System for Precision Farming Using Microcontroller

The study by Kurundkar et al. (2024), presents an automated precision irrigation system designed to optimize water usage and improve crop productions. The system uses IoT technologies like microcontrollers, sensors, and cloud platforms to ensure optimal water management is environmentally sound. The intelligent system not only provides automated and manual options but also reduces waste and labor costs which makes it a greener solution for farmers. Mobile app users will have a real-time monitoring and remote-control feature and thus will be able to utilize the system called the ThingSpeak IoT cloud. This idea helps to innovate old-fashioned irrigation methods.

The study implements the dual microcontroller architecture, which consists of an AT89c51 microcontroller for automation and ESP8266 for manual control and data transmission. This hardware redundancy enables the system to operate smoothly even in case of hardware failure making the system more reliable.

The system uses soil moisture sensor, rain sensor, and a DHT11 temperature and humidity sensor to collect real time environmental data. Based on pre-set threshold the system automates the water distribution which optimizes water usage avoiding over or under irrigation (Kurundkar et al. 2024). The use of rain sensor stops the irrigation during the rainfall preventing the crop damage and water logging. Farmers can not only remotely monitor the soil moisture, temperature and humidity but also switch between manual and automatic irrigation modes using a mobile app. Such an integration of IoT and cloud technology ensures convenience and flexibility for farmers to manage the system remotely.

The study emphasis on cost-effective design, utilizing affordable and easily available components like 9V water pump, basic sensors and other required electric modules. During the test, system performed remarkably. It significantly reduced the water wastage while maintaining optimal crop growth.

Limitations:

Though the system is innovative, it has some limitations. The system depends on pre-set moisture and temperature threshold which does not consider dynamic environmental changes or varying crop types. It also relies on Wi-Fi connectivity, which limits its applicability in rural regions with limited internet infrastructures.

## 11. Smart Farming using IoT: A Solution for Optimally Monitoring Farming Conditions

The study proposed by Doshi et al. (2019), focuses on using IoT to address inefficiencies in traditional irrigation approaches. The study exhibits a smart IoT-powered agriculture prototype consisting of ESP32 microcontrollers and sensors such as DHT11 and SI1145 as well as mobile applications which are useful in tracking environmental parameters, namely temperature, humidity, soil moisture, and UV index. The prototype has been designed in such a way that it can offer live data readings with notifications, so farmers can use this edition to

make the right choices based on the data. The main purpose of the research is to augment water savings, output of crops, and resource optimization through technology growth.

The proposed system uses ESP32 microcontroller, DHT11 temperature and humidity sensor and SI1145 UV and IR index sensor to monitor the environmental data. System provides a user-friendly interface through the Blynk mobile app for remote monitoring. The system also focuses on affordability, as its components were selected to be cost effective making it affordable for the small-scale agriculture and farmers. It also incorporates the sleep mode which further saves the energy further reducing the operational cost. The prototype uses LEDs and buzzers for localized notifications and the Blynk app to deliver the real-time data stream instead of just making sure that the farmers are immediately aware of the extreme situations such as low soil moisture and high UV levels (Doshi et al. 2019). Taking care of the soil, moisture, and UV levels is the electric vehicle to our planet health leading to maximum efficiency and minimum expenses. Besides that, farmers would be able to know the weather conditions in their farm region and through accuracy, success in agriculture is predictable to a point. Reducing chemical use means fewer greenhouse gases and cleaner soil and air conditions so, the project starts from the use of smart agriculture sensors. The model then provides insights into how the farm could expand by connecting several nodes using Bluetooth communication. The future direction of the project includes implementing artificial intelligence so the system can make decisions on its own and interfacing with drone technology that can produce a 3D visualization of the farmland.

Limitations:

Despite its significant strengths, it still has some limitations. The system heavily relies on stable Wi-Fi connectivity for optimal operation which may restrict its applicability in rural regions. Though the system automates the data collection process, it does not properly integrate the autonomous decision making by the system requiring manual farmer's intervention. Also, the farmers may need some training to use the Blynk app and interpret the data, this may limit the system adoption in technically backward regions.

## 12. Design of an Internet of Things (IoT) Based Smart Irrigation and Fertilization System Using Fuzzy Logic for Chili Plant

The paper by Pezöl, Adnan, and Tajjudin (2020) presents shows an innovative way of regulating watering and fertilizing for the intended chili plant. The authors state that chili plants are very touchy to soil moisture and pH levels during their development which leads to the requirement of exact management for better productivity. The system streamlined by the Internet of Things technology plus fuzzy logic controllers makes sure that even minute water and nutrient delivery with the assistance of a live-time monitor is achieved. Since chili plants serve as an important economic crop, the research focuses on automating and thus improving the traditional farming methods by the process of introducing modern technology.

The authors remarkably contribute to smart agriculture by inventing a system that is a combination of the IoT and fuzzy logic. The system is made up of sensors for moisture and pH

that are linked to some controllers (Arduino and Node MCU), enabling automatic water and nutrient delivery. The data is collected and displayed through the Blynk mobile application, giving the farmers real-time monitoring and control. Study presents a unique idea by using fuzzy logic controllers will be used to control the acidic, basic, and neutral solution flow based on the data from sensors, which will result in the precise application of irrigation and fertilization. The fuzzy rules that are programmed in MATLAB make it possible for the system to continuously adjust soil conditions so that they are at a peak level. Experiments and real applications in the field confirmed that the system is stable in providing healthy soil for the plants. The experimental program on chili plants for eight weeks has generated hopeful outcomes. Using specific irrigation and fertilization has proven to be more beneficial in terms of plant height and leaf count as compared to the conventional methods. This is a clear sign that the system can be used to boost production and at the same time save natural resources like water and fertilizers (Pezol et al., 2020).

#### Limitations:

The study exhibits certain limitations. Since the system is specifically tailored to chili plant, it is harder to implement it in broader crops types without significant modifications. Also the reliance on WiFi limits its range to about 6.5 meter making it inapplicable for larger agricultural fields. Besides the dependency on commercial platforms like Blynk may further introduce financial and technical challenge in rural areas.

### 13. Intelligent Urban Irrigation Systems: Saving Water and Maintaining Crop Yields

This paper is concerned about the agricultural water consumption, and it states that 70% of global water resources are being withdrawn for irrigation (Mason et al., 2019). These smart irrigation systems are recommended as an idea that helps to reduce water loss and environmental harm while at the same time achieving the desired crop production. The study principally examines urban agriculture through the application of intelligent irrigation systems in urban tomato gardens spread over nine diverse climatic regions in the U.S. With the help of the CROPWAT model, the researchers measure water consumption and yield in traditional and intelligent irrigation methods, pointing to the role of wireless soil moisture sensors and automated valves in better water use and irrigation scheduling.

The research has a significant relevance to the development of water-efficient urban farming. Mason et al. (2019) show that in sub-humid climates it got up to a 59% average of water saving due to smart irrigation without a change in crop yields. The CSIS Index is a new method of measuring irrigation efficiency in terms of both water and farmland food production. The research also highlights the ability of intelligent systems to increase and improve the use of rainwater for crops by 36% in sub humid climates. This ensures that the crops can make better use of rainfall and benefit from it. The research highlights some insights about the regional variability of intelligent irrigation systems. The system is effective in sub-humid regions while its performance is marginal in semi-arid-climates where the water saving is not that effective.

#### Limitations:

Though the research highlights several valuable insights, there are some limitations that need to be discussed. One of the limitations is assumption of same or uniform loam soil properties all across the simulated regions which does not consider the variability in urban agricultural settings (Mason et al., 2019). Also, the system does not address the cost associated with the installation and operation of the irrigation system which significantly hampers the adoption of the technologies in real world application.

## Summary

The above studies on smart irrigation system collectively highlight the significant progress in the use of AI and IoT in agriculture. The study demonstrates the potential of IoT combined with AI to optimize the water usage and improve the agricultural efficiency leading to higher crops production. Significant innovations include the use of AI prediction algorithms for rainfall, weather conditions, and use of low cost sensor integrations for data collections. Also the IoT based remote control and monitoring is a key highlight. The experimental results across various studies of such systems show high accuracy, reliability in communication, and reduction in water wastage. The findings show the adaptability of these technologies to different agricultural environments and settings. The study also focuses on the significance of real time data, modular system designs, and also the use of energy efficient communication networks like LoRa for remote communication. Overall, the study points out the various technological solutions and approaches in addressing water resource challenges in agriculture and global food security.

However, in context of my thesis project, various issues like financial limitations, time constraints, agricultural land constraints, and lack of powerful hardware for AI models need to be considered. Such limitations and constraints necessitate the use of simplified yet effective technologies. The available core components that are going to be used in this case are, ESP32, soil moisture sensor, temperature and humidity sensor, a simple water pump/motor, and a simple AI model able to predict irrigation needs based on the parameters like soil moisture, temperature, humidity, plant type and plant days (age). Despite the constraints, the project shall align well with the core findings and foundational principles and concepts of the aforementioned reviewed studies. While the relatively expensive and advanced technologies like LoRa network or hydroelectric-powered nodes may not be feasible, basic IoT communication and AI based decision making seems practical and impactful and hence shall be adapted in the project. The focus on real time monitoring of the environment and automation using the low cost and easily available components ensures that the project is cost effective and manageable. Having said that, the project still contributes to the goals of sustainable irrigation and water conservation. This approach also demonstrates the scalability and adaptability of smart irrigation solutions even within the constrained resources and other limitations. (Also the water usage statistics shall be calculated to give insights on water consumption by specific type of plants throughout its growth).



## Chapter 4 : Primary Research

### Objective of primary research

The main objective of the primary market research is to gather valuable insights on the practical challenges, market requirements, and also the opinions of the farmers or users regarding the smart irrigation system. By understanding their knowledge, experience, opinions and views on the given topic, this research aims to gather following details and insights:

- Study the irrigation methods currently being used and the challenges associated with them such as water wastage, inefficient resource utilization etc.
- Evaluate and assess the farmer's knowledge and awareness regarding smart technologies such as IoT and AI, and also their willingness to adopt such technologies in their daily life.
- Identify and evaluate the resource constraints such as financial, technical and infrastructural limitations specially in rural areas that may affect the adoption of smart irrigation system.
- Collect and analyze the user requirements and expectations on such systems such as affordability, reliability, ease of use and maintainability.
- Analyze the potential barriers to adoption of smart irrigation system like lack of technical knowledge, network connectivity issues, or other socio-cultural factors in rural areas.
- Explore and evaluate the user interests in the inclusion features like irrigation automation, water usage tracking, plant age monitoring etc in the system.

### Survey Questionnaire

An online google form was created consisting of about 15 to 16 related questions and was sent to the potential users to collect their responses. By evaluating and analyzing those collected responses from target audience, the research aims to validate the feasibility, relevance and acceptability of the smart irrigation system while also ensuring that the smart irrigation system effectively and efficiently addresses the needs and requirements of the farmers and helps in promoting sustainable farming practices.

| S.N | Question                                                                  | Objective of Question                                                                                                                                   |
|-----|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.  | Are you interested in making good income from you agricultural products ? | To know whether people want to increase their agricultural production and use it as a source of income and their likelihood to adopt such technologies. |
| 2.  | What is your primary role in agriculture ?                                | To classify the responses by their role in agriculture.                                                                                                 |
| 3.  | What type of crops do you primarily grow ?                                | To gather the data on diversity of crops and their compatibility with the system.                                                                       |

|     |                                                                                                            |                                                                                                                        |
|-----|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| 4.  | What is the size of your agricultural land ?                                                               | To assess the applicability and scalability of the system in various sized lands.                                      |
| 5.  | What irrigation method do you currently use ?                                                              | To evaluate the current irrigation methods in use, and identify the gaps that this system can fill.                    |
| 6.  | How often do you monitor soil moisture levels before irrigating ?                                          | To assess current irrigational methods' precision.                                                                     |
| 7.  | What challenges do you face with your current irrigation system ?                                          | To identify and evaluate the challenges faced by farmers in current irrigation methods.                                |
| 8.  | Are you familiar with the concept of smart irrigation systems ?                                            | To know the awareness of smart technologies in target users.                                                           |
| 9.  | Have you used any IoT-based devices in your farming operations ?                                           | To evaluate the current adoption and familiarity of such technologies.                                                 |
| 10. | What factors would influence your decision to adopt a smart irrigation system ?                            | To address the challenges and constraints hindering the adoption of smart irrigation system.                           |
| 11. | How much are you willing to spend in devices like smart irrigation systems that automate your irrigation ? | To evaluate the financial capacity of target users and their willingness to pay for such technologies.                 |
| 12. | How important is water conservation in your farming practices ?                                            | To evaluate farmer's concerns about the water conservation and it's influence in adopting smart irrigation system.     |
| 13. | Would you prefer an irrigation system that provides automated decision-making ?                            | To evaluate the demand and acceptability of irrigation system in the market.                                           |
| 14. | What features would you like in a smart irrigation system ?                                                | To collect user's preferences and expectations on the smart system to design it to meet the end user needs.            |
| 15. | What challenges do you foresee in adopting smart irrigation systems ?                                      | To evaluate the possible barriers in adoption of such technologies such as network connectivity, maintenance cost etc. |

|     |                                                                                                |                                                                                                          |
|-----|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| 16. | Would you be willing to participate in a trial or demonstration of a smart irrigation system ? | To know the people's interests in engaging in the trial programs and gaining experience with the system. |
|-----|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|

Table 2: Survey questionnaires with objectives

## Survey Response

Are you interested in making good income from you agricultural products ?

221 responses

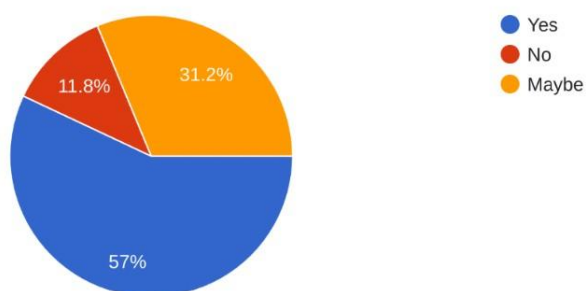


Figure 4: People's interest in making good income from agriculture

The above figure shows that the majority of 57% of the people are interested in making a good income from their agricultural products and 31.2% are unsure about it. This highlights the people's interest in earning from agriculture hence increasing the chance of adoption of technologies that can potentially increase the crop production.

What is your primary role in agriculture?

222 responses

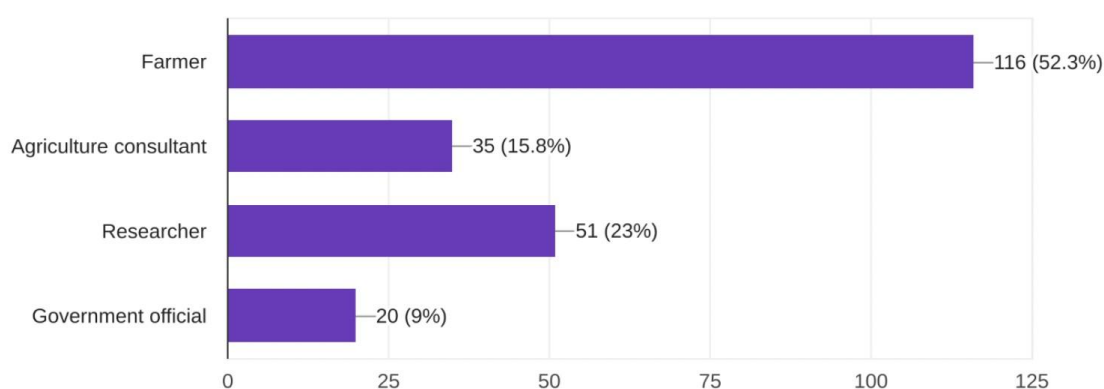


Figure 5: People's role in agriculture

The above figure shows the majority i.e. 52% of respondents are farmers which suggests that the research has reached the target audiences capturing their feedback. Involvement of

researchers and agriculture consultants provides extra perspectives to the research on the adoption and viability of the system.

What type of crops do you primarily grow

222 responses

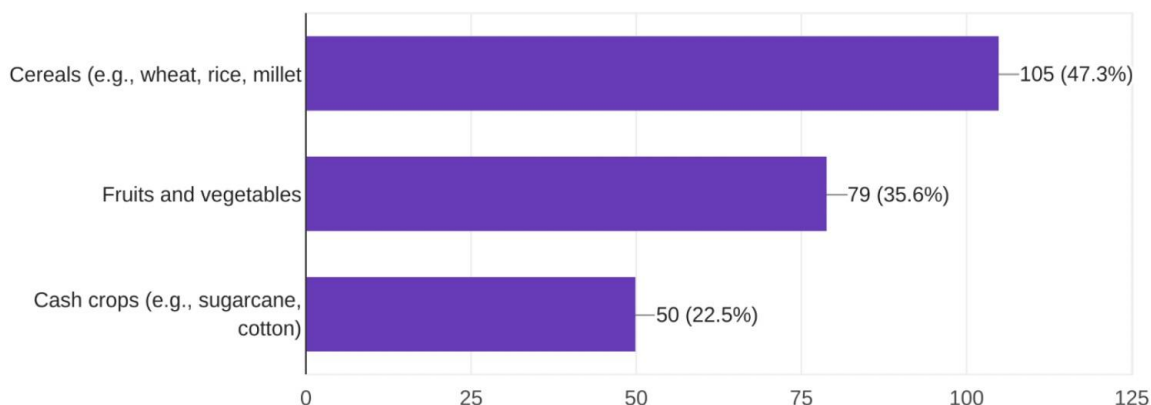


Figure 6: Types of crops grown by the respondents

The above figure shows that the cereals, fruits and vegetables are the main crops that are being grown. This shows that the irrigation needs vary significantly across various crop types and hence requires a system that can adapt to diverse crop types and their water requirements.

What is the size of your agricultural land?

221 responses

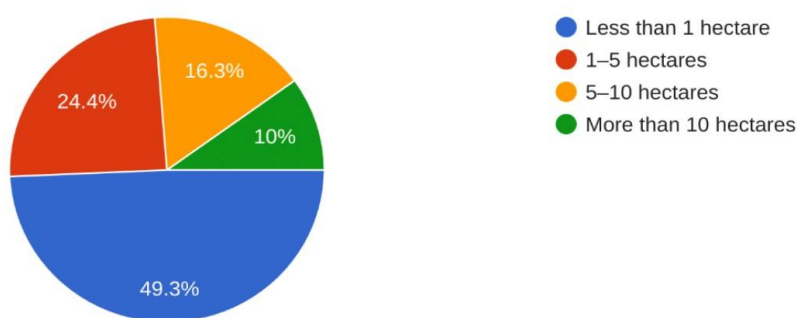


Figure 7: Size of agricultural lands among respondents

The above figure shows that, most of the target audiences have below 1 hectare and 24% of them have 1 to 5 hectare of land indicating the need of small scale, cost effective and scalable precision irrigation system.

### What irrigation method do you currently use?

222 responses

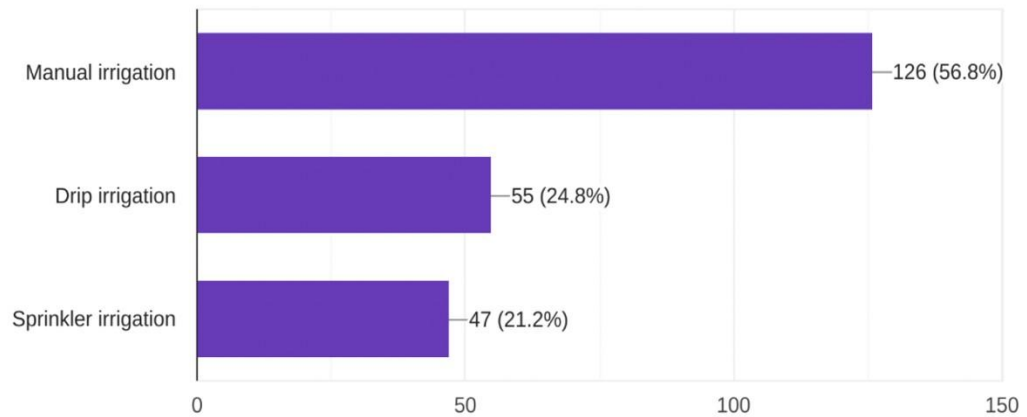


Figure :8Existing irrigation methods in use

The above figure shows that the majority of farmers use manual irrigation which is inefficient and it can be improved by the precision irrigation system. Smart irrigation systems can optimize the water use and also reduces the manual labor. The system seems feasible and adoption rate is found to be higher from the above response.

### How often do you monitor soil moisture levels before irrigating?

222 responses

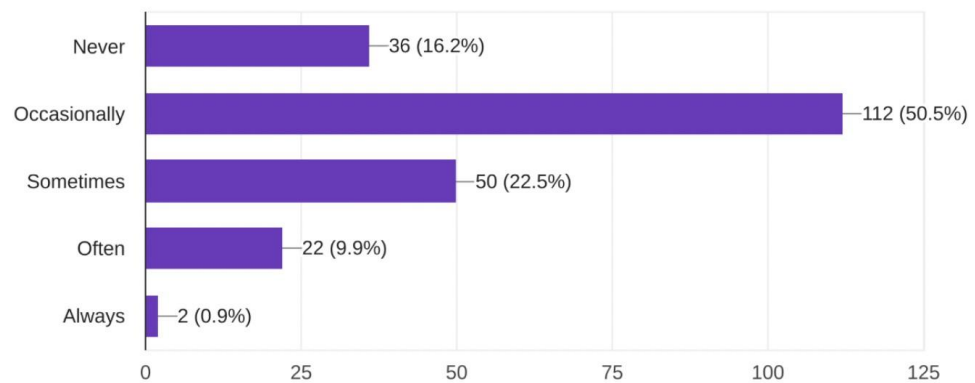


Figure 9:Frequency of soil moisture level monitoring

The above figure of responses shows that the majority of people only occasionally and 16% never measure the soil moisture level before irrigating, this could lead to over or under irrigation leading to water wastage or decreased productivity. This highlights the need of smart irrigation system that automates the irrigation considering soil moisture.

### What challenges do you face with your current irrigation system?

222 responses

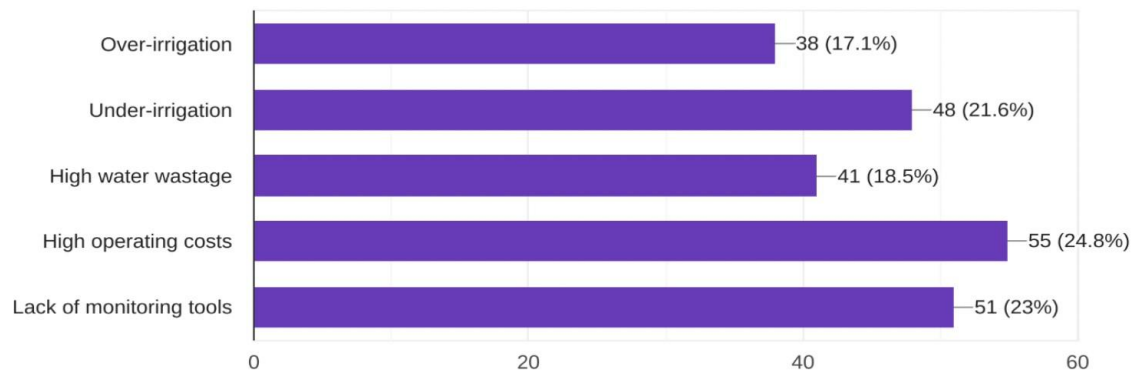


Figure 10: Challenges faced in current irrigation system

The above figure shows that the people face several challenges like over irrigation, under irrigation, water wastage and also the lack of monitoring tools which highlights the need of efficient smart irrigation system which addresses these challenges.

### Are you familiar with the concept of smart irrigation systems?

222 responses

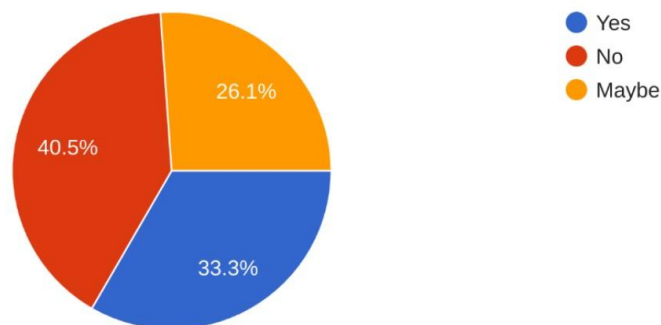


Figure 11: People's familiarity with current smart irrigation system

The above figure shows that majority of people are unfamiliar or unsure about the smart irrigation system. This indicates the system requires trial phases to demonstrate it's working and give hands on experience to the people. Also it requires user friendly system as people may lack technical knowledge.

## Have you used any IoT-based devices in your farming operations?

222 responses

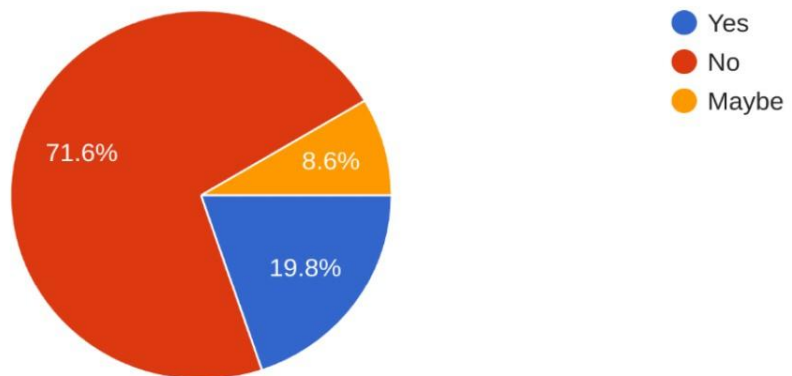


Figure 12: People using IoT devices in the past

The above response indicates that 71% of the people never used IoT devices. While this shows the knowledge gap which required some training, this also shows that these people can be the early adapters for the trial of the prototype system. If the system is reliable and efficient, it has high chance of being adopted.

## What factors would influence your decision to adopt a smart irrigation system?

222 responses

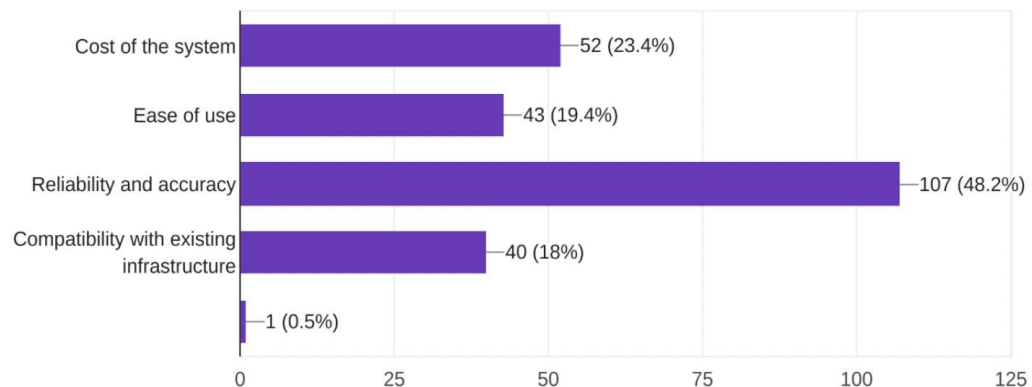


Figure 13: Factors affecting the adoption of smart irrigation system

The above figures show that among several factors influencing the decision to adopt smart irrigation system, most of them are more concerned about the reliability, accuracy and cost of the system. Indicating that the system must be cost effective, reliable and efficient while also maintaining the ease of use.

How much are you willing to spend in devices like smart irrigation systems that automate your irrigation ?

222 responses

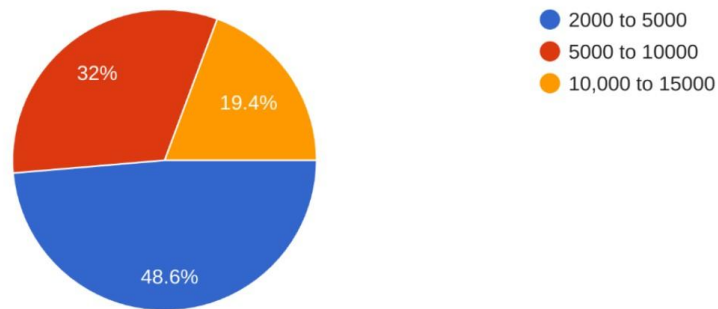


Figure 14: Expense preferences of respondents

The above figure shows that majority of people are willing to pay below Rs 10,000. So the system must be cost effective utilizing cheaper and effective IoT components.

How important is water conservation in your farming practices?

222 responses

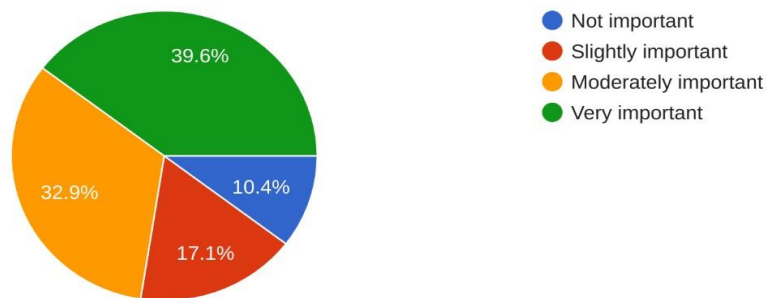


Figure 15: Importance of water conservation among people

Above figure shows that above 60% of respondents rate water conservation as very or moderately important. This highlights people's interests in saving the water. This increases the adoption rate of the smart irrigation system.

Would you prefer an irrigation system that provides automated decision-making?

222 responses

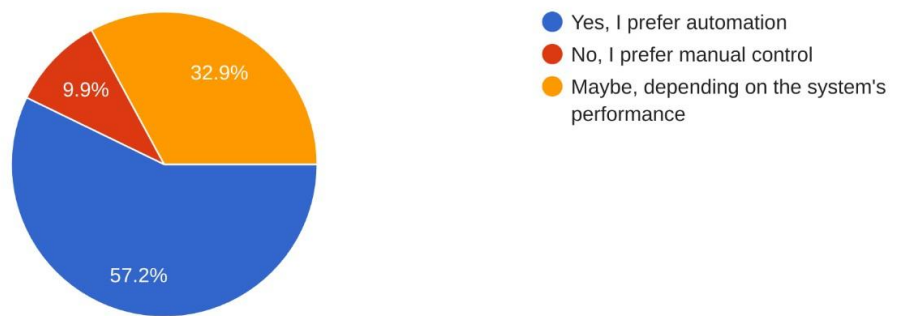


Figure 16: Preferences of automated systems

Above figure shows that most of the people prefer an automated irrigation system, showing the feasibility and acceptability of the smart irrigation system.

What features would you like in a smart irrigation system?

222 responses

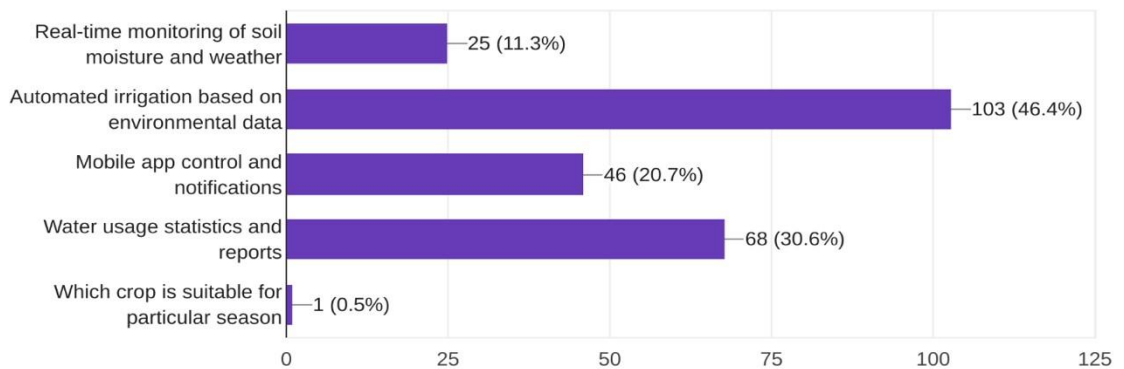


Figure 17: Preferred features in smart irrigation system

Above diagram highlights the people's preferred features in a smart irrigation system. It is clear that the system with automated decision-making based on real-time environmental data is preferred. Features like water usage statistics are also preferred by many people.

### What challenges do you foresee in adopting smart irrigation systems?

221 responses

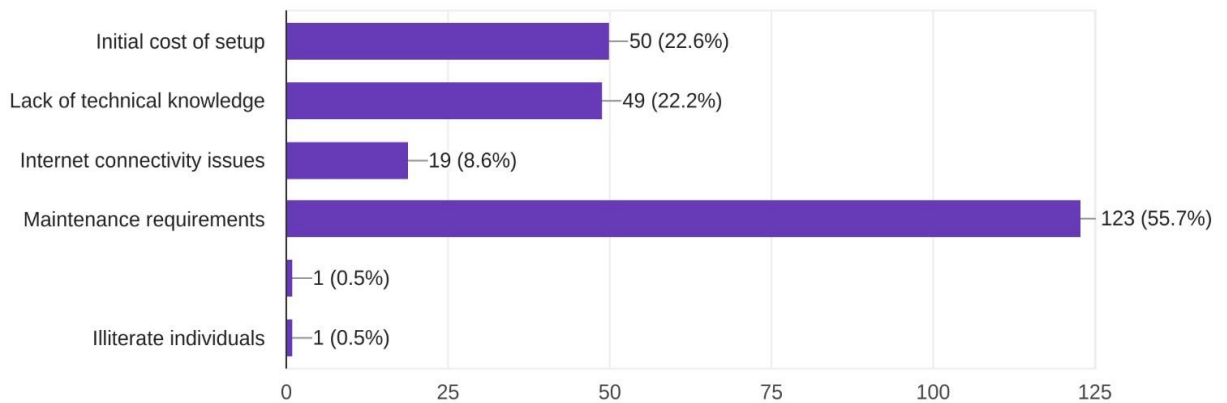


Figure 18: Challenges in adopting smart irrigation system

The above figure shows that the main challenges in adopting the smart irrigation system are the maintenance requirements, setup cost and technical knowledge. This indicates that the system must be durable requiring less maintenance, it should also be cost effective and user friendly to address challenges in rural areas.

### Would you be willing to participate in a trial or demonstration of a smart irrigation system?

222 responses

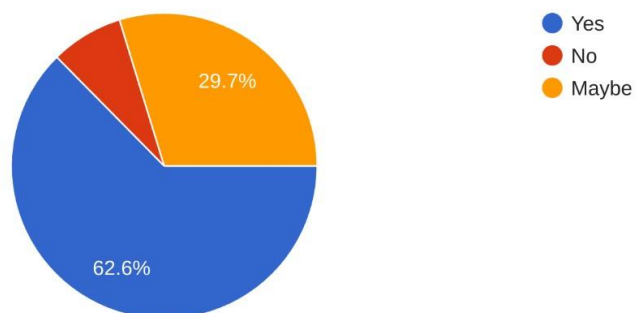


Figure 19: Willingness to participate in system trial demonstration

The above figure shows that most of the respondents are willing to take part in the trial or demonstration of the smart irrigation system. This highlights people's interest in trying new technologies. Involving in trial gives people hands on experience with the system, people can also assess the system's performance and reliability. This increases the chances of adoption of system if it aligns with their expectations.

## Data collection and analysis

The whole project's success depends upon the systematic and structured approach to data collection and analyses for developing a practical and robust smart irrigation system. A combination of both primary and secondary data sources was utilized. Insights from literature review and from market surveys were taken.

### Data Collections

#### Primary Data Collection:

The survey was done via google forms, including a choice-based questionnaire to extract information and ideas from people indulged in agriculture. The motive of the survey is to apprehend the prevailing irrigation techniques in practice, challenges faced, willingness of people to adopt smart technologies, budget limitations, and the features they expect in smart irrigation systems. This survey has helped in identifying the feasibility challenges and user preferences through qualitative and quantitative data.

#### Secondary Data Collection:

A detailed and exhaustive literature review of the peer and relevant academic sources has been carried out to understand the knowledge base that exists in the market regarding the existing technologies, their methodologies, approaches, and limitations. Some published research paper and case studies were studied and evaluated to gain insights into the current trends in the arena of IoT, AI, and sensor-based precision irrigation systems.

#### AI Model Training Data Collection:

Preparing the dataset for developing an AI model requires training and validation of the predictions for irrigation requirements. Datasets can be obtained from sources like Kaggle, GitHub, and academic repositories. These datasets generally contain parameters such as soil moisture, temperature, humidity, and the need for irrigation. Some of them provide this data with additional parameters like crop type, age of the crops, and seasonal changes. Some of the retrieved data will need to be preprocessed to include normalizations, handling of missing values, and feature selection in order to conform with the lightweight AI model used here in the project.

## Data Analysis

#### Quantitative Data Analysis:

The results of the survey were analyzed statistically to find some patterns and preferences. Some of the metrics include what percentage of farmers face particular challenges, how many are willing to adopt IoT systems in place, and the budget constraints. These insights led to designing choices made with cheaper and effective components and simplifying the usability of the system.

#### Qualitative Data Analysis:

The responses were analyzed and grouped to identify key insights through the survey, such as barriers of adoption, desired features, and trust in new technologies. A literature review has helped also to deepen the understanding of the gap in current smart irrigation solutions and guided the architecture.

### Summary of research

The survey highlights the challenges and opportunities in the design of a smart irrigation system for farmers in poor areas, which is one of the most important problems to be solved. The main problems with the irrigation systems that are used today are that they are too expensive, power usage is inefficient and monitoring is not available, thus showing the need for cheaper, more accurate, and more efficient solutions. However, smart irrigation systems are not a widely known and understood topic, the respondents show their preferences of real-time monitoring, automated decision-making, and smart energy efficiency systems as their most popular features. It is known that most participants would not spend more than NRs 1000 which not only highlights the need to develop low-cost and accessible systems, but, also, implies that cost remains a critical factor. While the technical knowledge gaps and initial costs are adoption barriers, the educational support and trial programs are the means to increase awareness and system acceptance. Even though the majority of respondents do not have any prior experience with IoT devices, their receptivity to automation and willingness to participate in trials give a perfect opportunity to show how the system can be of benefit, and hence, help in the adoption of the system. A smart irrigation system that is cost effective, user friendly, and water-saving can address the challenges in agricultural irrigation and promote sustainable agricultural practices.

## Chapter 5 : Artifact planning

### Requirement analysis

#### Functional Requirements

| S.N. | Requirement                                                                                                                                           | Priority |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1    | The system must be able to accurately collect and process data from soil moisture sensor, temperature and humidity sensor and real time clock module. | MUST     |
| 2    | The microcontroller must be able to send and receive the data to and from the server.                                                                 | MUST     |
| 3    | The system must be able to utilize the AI model deployed in the server to make irrigation decisions based on sensor readings.                         | MUST     |
| 4    | The system must be able to start or stop the water pump based on AI model's prediction.                                                               | MUST     |
| 5    | The system must be able to keep track of crop age via it's real time clock module.                                                                    | MUST     |
| 6    | The system should be able to calculate and store the total amount of water consumption by each crops throughout their life span until harvesting.     | SHOULD   |
| 7    | The system must provide intuitive dashboard via web app for remote monitoring of environmental parameters and irrigation status.                      | MUST     |
| 8    | The system should provide option for manual override via web app to manually activate or deactivate the irrigation.                                   | SHOULD   |
| 9    | The system may store and analyze the historical weather and rainfall data to make better predictions.                                                 | COULD    |

Table 3:Functional requirements of the project

#### Non-function Requirements

| S.N. | Requirement                                                                                 | Priority |
|------|---------------------------------------------------------------------------------------------|----------|
| 1    | The server should be able to simultaneously handle multiple systems' connection.            | MUST     |
| 2    | The system should use cost effective components to ensure applicability in rural areas.     | MUST     |
| 3    | The system should be able to handle fluctuating environmental conditions.                   | MUST     |
| 4    | The system should be scalable to add more sensors to adapt to various agricultural settings | SHOULD   |
| 5    | The system should be power efficient, it should use sleep mode to save energy.              | SHOULD   |

|   |                                                                                             |        |
|---|---------------------------------------------------------------------------------------------|--------|
| 6 | The system should be able function consistently without failures for a long time.           | MUST   |
| 7 | The AI model should be lightweight to ensure its smooth operation even in low end hardware. | SHOULD |
| 8 | The storage and transfer of the collected data should be secure.                            | COULD  |

Table 4:Non-functional requirements of the project

### Usability Requirements

| S.N. | Requirement                                                                                                                                                                                    | Priority |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1    | The user interface should be intuitive even for farmers with less technical experience.                                                                                                        | MUST     |
| 2    | The system should have easy and intuitive user manual for it's operation.                                                                                                                      | MUST     |
| 3    | The user interface should include visual tools to represent the irrigation schedules along with current system status which includes environmental conditions, irrigation status, crop age etc | SHOULD   |
| 4    | The system should be able to handle the network connectivity issues and sync data with the server when the network is available.                                                               | SHOULD   |
| 5    | The system should support multiple local language to increase it's acceptance and accessibility.                                                                                               | SHOULD   |
| 6    | The system should be able function consistently without failures for a long time.                                                                                                              | COULD    |
| 7    | The system should be durable with minimal maintenance requirement.                                                                                                                             | SHOULD   |

Table 5:Usability requirements of the project

### Software Requirement

- Vscode with PlatformIO IDE extension.
- Arduino and esp32 device drivers for uploading sketch to the microcontrollers.
- Python 3 along with other libraries like numpy, panda etc.
- AI and ML libraries like scikit-learn, tensorflow, pytorch etc.

### Hardware requirement

#### Development requirement

- Nvidia GTX 1050ti or later GPU.
- Intel core I5 8th gen, amd ryzen 3 3200 or later CPU

- ESP32 microcontroller
- Temperature and humidity sensor
- Soil moisture sensor
- Real time clock module
- Water pump/motor
- WiFi with network connectivity

### Deployment Requirement

- Cloud server with minimal GPU access to smoothly run the AI model
- ESP32 microcontroller
- Soil moisture sensor
- Real time clock module
- Water pump/motor
- WiFi with network connectivity

## System design

### Use case

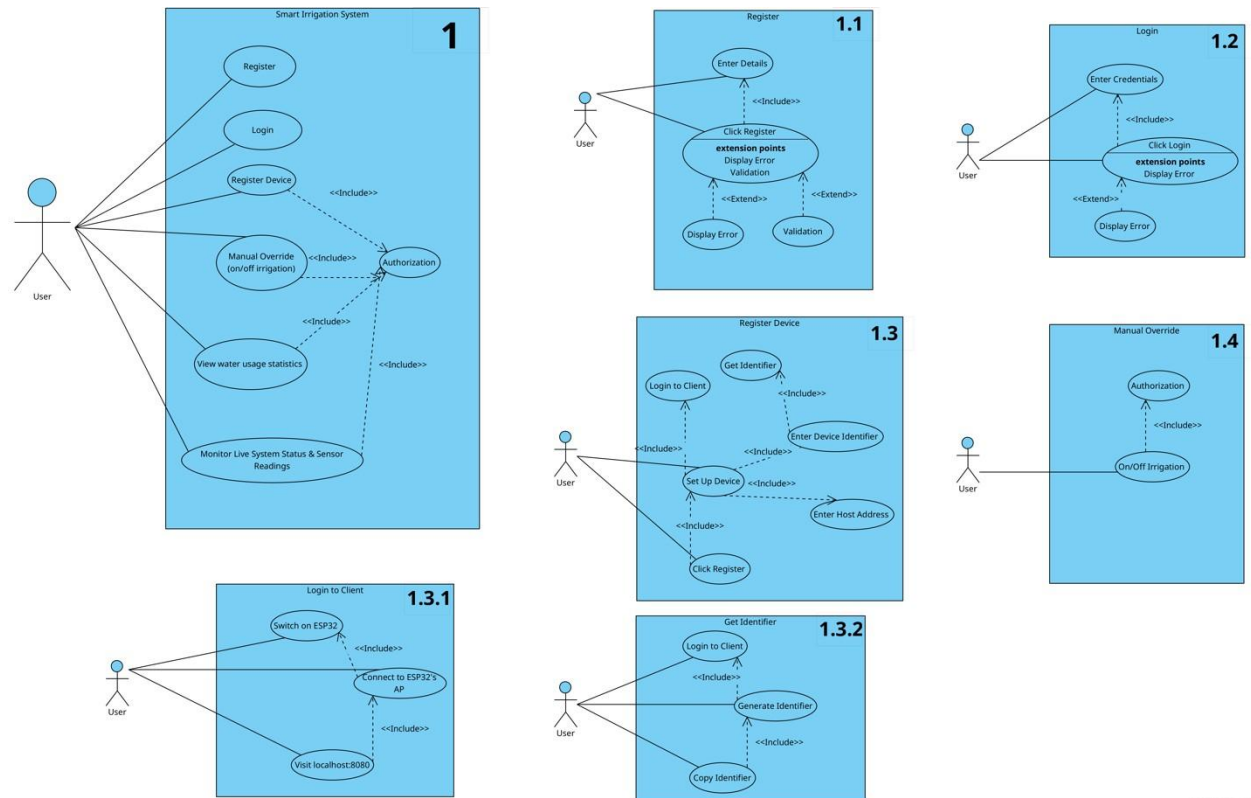


Figure 20: Use case diagram of the project

## DB schema

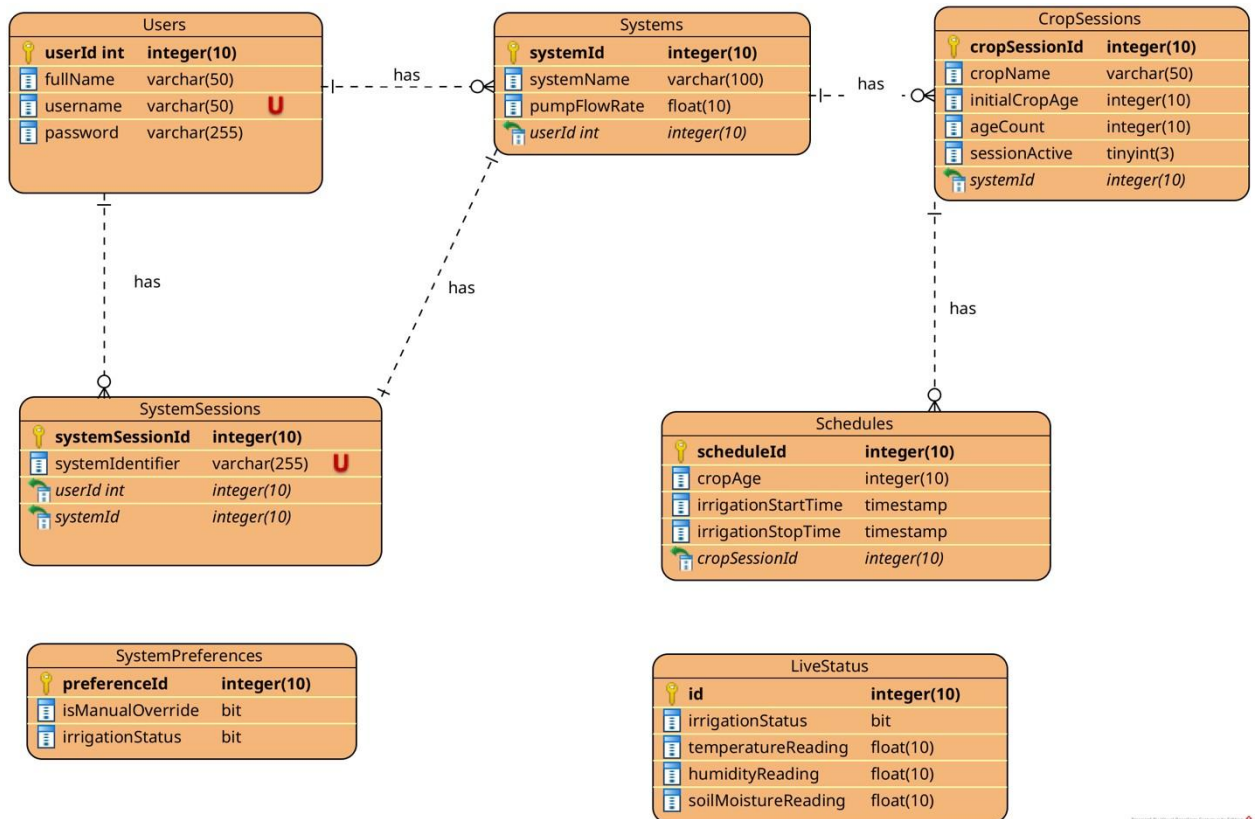


Figure 21:ER Diagram of the project

## Chapter 6 : Testing and Evaluation Strategy

### 6.1 Testing Strategy

Accuracy, functionality, and reliability of sensors, AI model testing, and overall performance of the smart irrigation system shall be tested in the following manner.

#### 1. Unit Testing

Every component that is used, such as the ESP32 microcontroller, soil moisture sensor, temperature and humidity sensor, and real-time clock module, will be first tested individually to ensure correct functionality. The sensor readings will be compared with known values for testing on accuracy.

#### 2. Integration Testing

Integration testing shall be done once the individual components are tested and validated. Integration test shall ensure that the components are integrated correctly and there is seamless communication between sensors, microcontrollers, and the server inclusive of AI model. The integration of a web app and the feature of manual override via the web app shall also be tested and validated.

#### 3. Functional Testing

It has to be tested within a controlled environment to ensure that the system operates just as expected. It should collect surroundings and soil data and transmit it to the server, thereby receiving instructions from the server to stop or start irrigation. The test to be conducted should cover the functionality when the soil is too dry; the pump needs to turn on, and if it is wet, then the pump needs to turn off according to the prediction made by the AI model based on varying environmental parameters. The tests will also verify a manual override via web app.

#### 4. Performance Testing

The system is required to be exposed under a controlled environment with fluctuating temperature, soil moisture, and connectivity disruptions.

### 6.2 Evaluation Strategy

The evaluation strategy focuses basically on the validation of the efficiency, effectiveness, and impact of the system in terms of water conservation and crop production.

#### 1. Quantitative Metrics

- Compare the water use by the smart irrigation system with that of traditional irrigation methods and determine the water-use reduction. Literature states that the precision irrigation systems reduce the water usage up to 46% in agriculture (Lakshmi et al., 2023)..

- Compare the production of crops in the field with smart irrigation systems with that of the field that is manually irrigated.
- Analyze and evaluate the power consumption of the system to ensure that it is sustainable.

## 2. Qualitative Metrics

- Collect farmers' views and opinions with regard to the adaptability, reliability, and usability of the system to changes in environmental conditions. Such feedback helps in further refinement and enhancement of the smart irrigation system.
- The evaluation and assessment of the system's ability to face real-world challenges, such as environmental conditions and connectivity issues.

## 3. Scalability and Adaptability

The system's flexibility would then have to be judged for scale-up on larger farms or various conditions of agriculture, including soil and crop requirements. Secondly, this will also include an AI model that has been trained on an extensive dataset of different crops and their water requirements corresponding to prevailing environmental conditions, which enhances its adaptability, while the application of remote monitoring increases its flexibility.

## Chapter 7 : Development

The development of Smart Irrigation System included the IoT and AI to automate an irrigation solution based on data driven approach. This part outlines the methodology that was used, tools and technologies that were employed, and the process of implementation in details.

This system was designed with the ability of soil moisture monitoring, temperature and humidity monitoring, and data on crop type and crop age to ensure better irrigation plans. The overall development process included hardware integration, software development, AI model training, and real-time monitoring features. The following sections contain a detailed discussion on the technology stack and the methodology chosen.

### 7.1 Methodology

The project employed the Agile methodology in software development, thus securing the possibility to use a flexible method and a series of changes for the entire process. An Agile strategy was utilized that made it possible to incorporate changes from feedback as the project was underway, as well as to build step by step and respond to challenges encountered during implementation.

Effectiveness of Agile Methodology

- Flexibility: Enabled iterative improvements based on testing and supervisor feedback.
- Continuous Integration: Allowed seamless hardware and software integration.
- Incremental Development: Ensured timely feature additions and improvements.
- Collaboration: Helped in regular supervisor meeting for problem solving then implementing the required changes.

### 7.2 Tools and Technologies Used

A combination of hardware and software technologies was used to ensure efficiency and reliability. The most important elements are:

**Hardware:**

ESP32 Microcontroller: Acts as the central processing unit for sensor data receiving and sending.

**Sensors:**

Soil Moisture Sensor: Measures soil water content to determine irrigation needs.

Temperature and Humidity Sensor (DHT11): Provides the environmental conditions i.e. temperature and humidity readings.

Water Pump and Motor driver (L293D): Controls irrigation based on AI-based decisions.

**Software and Development Tools:**

VSCode PlatformIO: To program the ESP32 microcontroller.

Python & Random Forest: For creating AI models and processing data.

NodeJS and Express: Back-end framework for real-time monitoring and API development.

Python Http Server: Another backend for deploying AI model.

Random Forest: Machine learning libraries for training the irrigation prediction model.

### 7.3 Risks Encountered

Though the risks related to the proper functioning and development of the irrigation system were analysed and also their mitigation strategies were discussed in detail in Risk Table part of the document. Thus those risks were kept in mind and carefully considered while developing and implementing the system, as a result those potential hindrances were avoided.

However some other unforeseen risks were encountered during the development phase of the system. Some of the major risks encountered are:

- **Hardware components failure**
- During the initial phase of development while the unit testing of each components was being conducted to ensure they function properly, the L293D motor driver module did not work properly and as a result the system was unable to control the water pump motor. I was unable to figure out what was really causing the problem, and after supervisor meeting, it was concluded that the hardware itself was defective. Then I replaced it with the working one.
- **Lack of low level knowledge**
- The main hurdle encountered during the development phase was the lack of low level understanding of how the microcontrollers work and also the fundamentals of C++ programming language.
- Throughout the college life every assignments were done using the high level languages like Java, python, JavaScript etc, which would automatically take care of the memory management, memory leakage, and also we barely had to consider about the underlying hardware components. Everything used to be taken care by the runtime environments, garbage collectors, and the operating system itself. However in case of microcontrollers programming, there's no operating system and also our program directly interacts with the hardware without any layers in the middle.
- I did not consider these things initially, as a result even though the program was working correctly and everything was right, the device would crash after a short time of operating. After debugging it was found that, it was crashing due to Stack Overflow exception.

- I consulted my supervisor, and he provided with the insights that I need to manually create and destroy the manually allocated memory in order to prevent memory leakage which was causing the stack overflow exception.

The aforementioned risks significantly delayed the development process, nonetheless they were solved gradually as I gained more knowledge on the subject matter and also the repeated supervisor meetings have become greatly helpful.

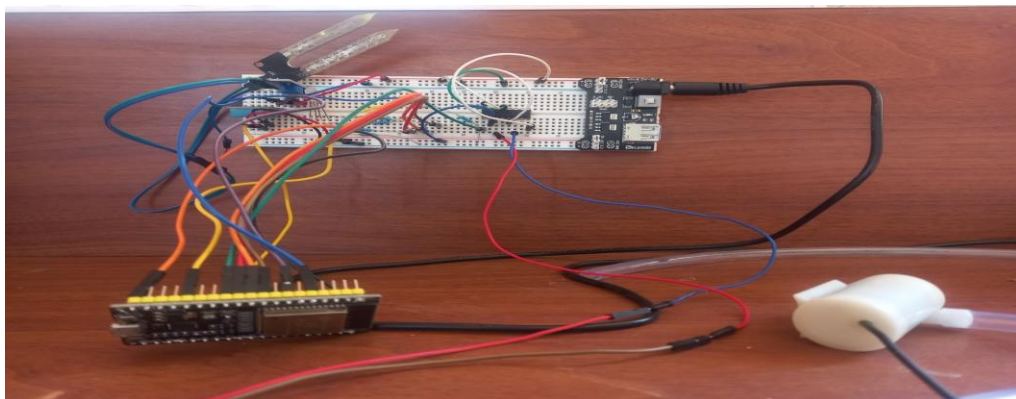
## 7. 4 Implementation Process

The system was developed in several key phases which are listed below:

Hardware Setup and Sensor Calibration:

- Individual Sensor readings were tested and calibrated for effective data collection.
- Microcontroller programming was done using VSCode PlatformIO.

First of all, individual components and sensors like temperature and humidity, soil moisture sensor and water pump with L293D motor drivers were connected to the microcontroller and tested if they work. Then the actual system programming was started.



*Figure 22:Final device after hardware components integration*

The above figure shows how the final IoT device or the irrigation system looks like once every sensors and microcontroller are assembled together.

**Data Collection and Preprocessing:**

There were numerous datasets available and in huge quantity but none of them contained the actual dataset required for my AI model. Mostly the datasets only contained, the columns like (temperature, humidity, soil moisture, irrigation status). However my system required two more columns i.e. (plant type and plant age).

This made the search for dataset really tough. Finally one dataset was found in kaggle but it only contained limited crop types and also the quantity wasn't huge. However the dataset was sufficiently enough for my purpose, as the system mainly focused on small scale farm and resource constrained environment, only supporting limited crops wasn't really a problem.

Since the dataset was available in kaggle, it did not require significant preprocessing, however minor label encoding was applied.

```
raw_data.sample(frac=1)
```

|     | CropType       | CropDays | SoilMoisture | temperature | Humidity | Irrigation |
|-----|----------------|----------|--------------|-------------|----------|------------|
| 300 | Potato         | 58       | 136          | 16          | 68       | 0          |
| 497 | Coffee         | 95       | 210          | 23          | 17       | 0          |
| 376 | Sugarcane      | 56       | 339          | 18          | 69       | 0          |
| 27  | Wheat          | 37       | 700          | 30          | 22       | 1          |
| 319 | Potato         | 94       | 330          | 16          | 68       | 0          |
| ... | ...            | ...      | ...          | ...         | ...      | ...        |
| 336 | Pulse          | 16       | 650          | 30          | 18       | 1          |
| 101 | Groundnuts     | 47       | 450          | 27          | 17       | 0          |
| 226 | Paddy          | 10       | 147          | 26          | 13       | 0          |
| 146 | Garden Flowers | 50       | 320          | 18          | 80       | 0          |
| 12  | Wheat          | 19       | 649          | 29          | 61       | 0          |

501 rows × 6 columns

Figure 23:Raw dataset

|     | CropType | CropDays | SoilMoisture | temperature | Humidity |
|-----|----------|----------|--------------|-------------|----------|
| 362 | 7        | 29       | 186          | 18          | 79       |
| 455 | 0        | 11       | 456          | 27          | 16       |
| 370 | 7        | 45       | 129          | 18          | 70       |
| 408 | 7        | 120      | 189          | 25          | 50       |
| 335 | 6        | 13       | 329          | 28          | 29       |
| ... | ...      | ...      | ...          | ...         | ...      |
| 190 | 3        | 39       | 440          | 34          | 19       |
| 128 | 1        | 1        | 400          | 24          | 70       |
| 111 | 2        | 66       | 780          | 22          | 21       |
| 419 | 7        | 142      | 689          | 32          | 38       |
| 44  | 8        | 77       | 450          | 21          | 14       |

Figure 24: :Dataset after label encoding

### AI Model Training and Optimization:

Once the data was collected from kaggle, various machine learning algorithms were experimented and trained on the same data. Finally the best model was selected. Various criteria like model accuracy, efficiency, prediction time and model size (or total numbers of parameters) were considered while selecting the final model to use in production.

| S.N | Model                        | Hyperparameters                                                                                                            | Accuracy % |
|-----|------------------------------|----------------------------------------------------------------------------------------------------------------------------|------------|
| 1   | MLP Classifier               | Hidden layer size: (10, )<br>learning rate: adaptive<br>max iter: 100 random<br>state: 42<br>n_iter_no_change: 10          | 88.11      |
| 2   | Sequential Keras, tensorflow | Layers: Dense(5, activation: relu), Dense(5, activation: tanh), Dense(5, activation: sigmoid)<br>loss: binary_crossentropy | 91.09      |
| 3   | Logistic Regression          | Max iter: 100 solver: lbfgs<br>class weight: balanced<br>optimizer: Adam                                                   | 91         |
| 4   | Random Forest                | n_estimators: 15<br>max depth: 8<br>random state: 0                                                                        | 98.02      |

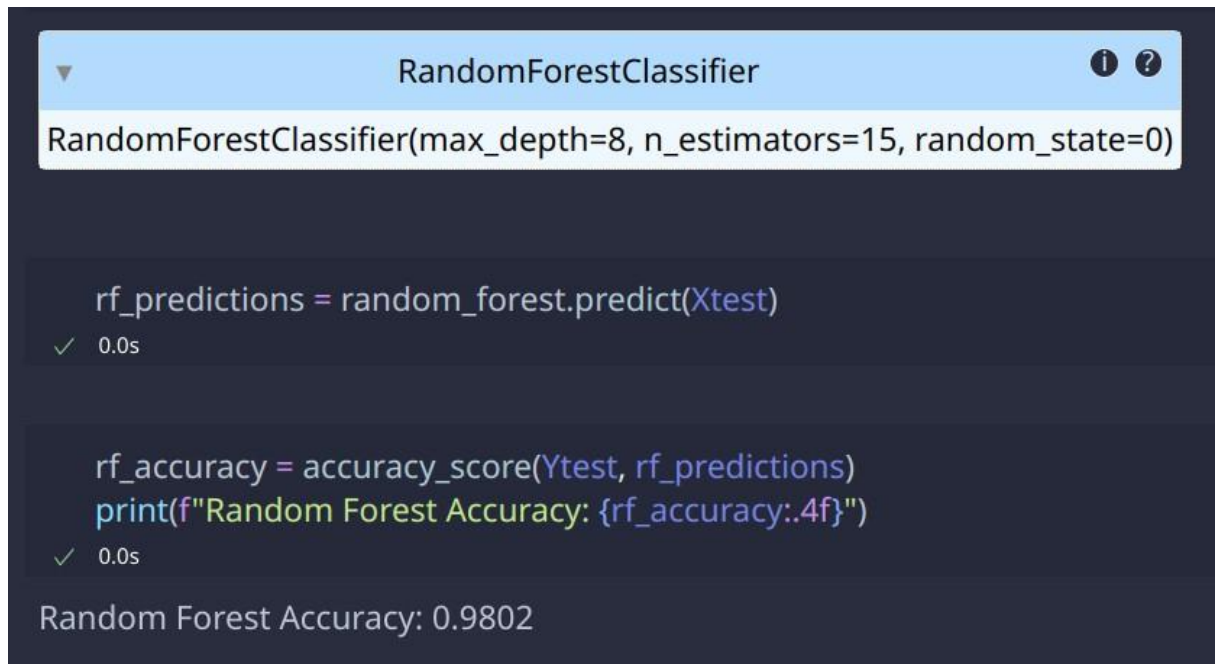
Table 6: AI Model Training and Optimization

<< Refer to appendix for other models confusion matrix >>

Figure 38: MLP Classifier: loss curve,

Figure 39: MLP Classifier: Confusion Matrix, Figure 40: Keras: model accuracy,

Figure 41: Logistic Regression: confusion matrix, Figure 42: Random Forest: confusion matrix)  
>>



```
RandomForestClassifier(max_depth=8, n_estimators=15, random_state=0)

rf_predictions = random_forest.predict(Xtest)
✓ 0.0s

rf_accuracy = accuracy_score(Ytest, rf_predictions)
print(f"Random Forest Accuracy: {rf_accuracy:.4f}")
✓ 0.0s

Random Forest Accuracy: 0.9802
```

Figure 25:Random Forest: model parameters and accuracy

Since the random forest model only uses 15 n\_estimators and maximum decision tree depth of just 8 levels, this makes the random forest model really lightweight and also decrease the prediction time because of less numbers of decision trees. This decreases the computational cost while still maintaining the decent accuracy score of 98%. So, it is decided that the random forest classifier model shall be used for the purpose of irrigation prediction in this project.

#### Backend and API Development:

The main backend server to handle the direct connection with the IoT device is written using nodejs. The server uses MySQL database for storing and retrieving data and Express library for building REST api. The IoT devices is controlled and monitored via this server.

For deploying the AI model, another python http server is built, all it does is handles one single REST api which receives the environmental data like temperature, humidity, moisture, plant type and plant age, then makes the prediction whether to irrigate or not. This server is deployed on the same machine where the nodejs server is deployed. The nodejs server sends request to the python server for making irrigation related decisions.

#### Frontend Development and Real-time Monitoring:

A web based dashboard is created using Reactjs and material UI for modern look and feel. The interface is kept simple and intuitive for easy navigation and ease of access. However users may need some guidance at first.

The user can monitor and control the system from this dashboard. User can view the live status of the farmland i.e. temperature, humidity and soil moisture as well as the current irrigation status. User can also manually control whether to irrigate or not by using the manual

override feature given in the dashboard. Besides, user can view the irrigation schedules and also the overall water consumption graph by the particular plant throughout it's lifetime.

| Farmland System   |                         |  |
|-------------------|-------------------------|--|
| Crop              | Pulse                   |  |
| Age               | 50 Days                 |  |
| Temperature       | 18 °C                   |  |
| Humidity          | 48 %                    |  |
| Moisture          | 1023                    |  |
| Irrigation Status | OFF                     |  |
| Last Updated      | 02/17/2025, 08:41:56 AM |  |

| Irrigation Schedules |                         |                         |
|----------------------|-------------------------|-------------------------|
| Crop Age (Days)      | Start Time              | Stop Time               |
| 49                   | 02/17/2025, 11:32:40 AM | 02/17/2025, 11:34:47 AM |
| 49                   | 02/17/2025, 08:38:55 AM | 02/17/2025, 08:39:59 AM |
| 49                   | 02/17/2025, 08:29:05 AM | 02/17/2025, 08:29:30 AM |
| 49                   | 02/17/2025, 08:24:06 AM | 02/17/2025, 08:27:24 AM |
| 49                   | 02/17/2025, 08:01:41 AM | 02/17/2025, 08:02:26 AM |

Figure 26: System Live Status and recent irrigation schedules

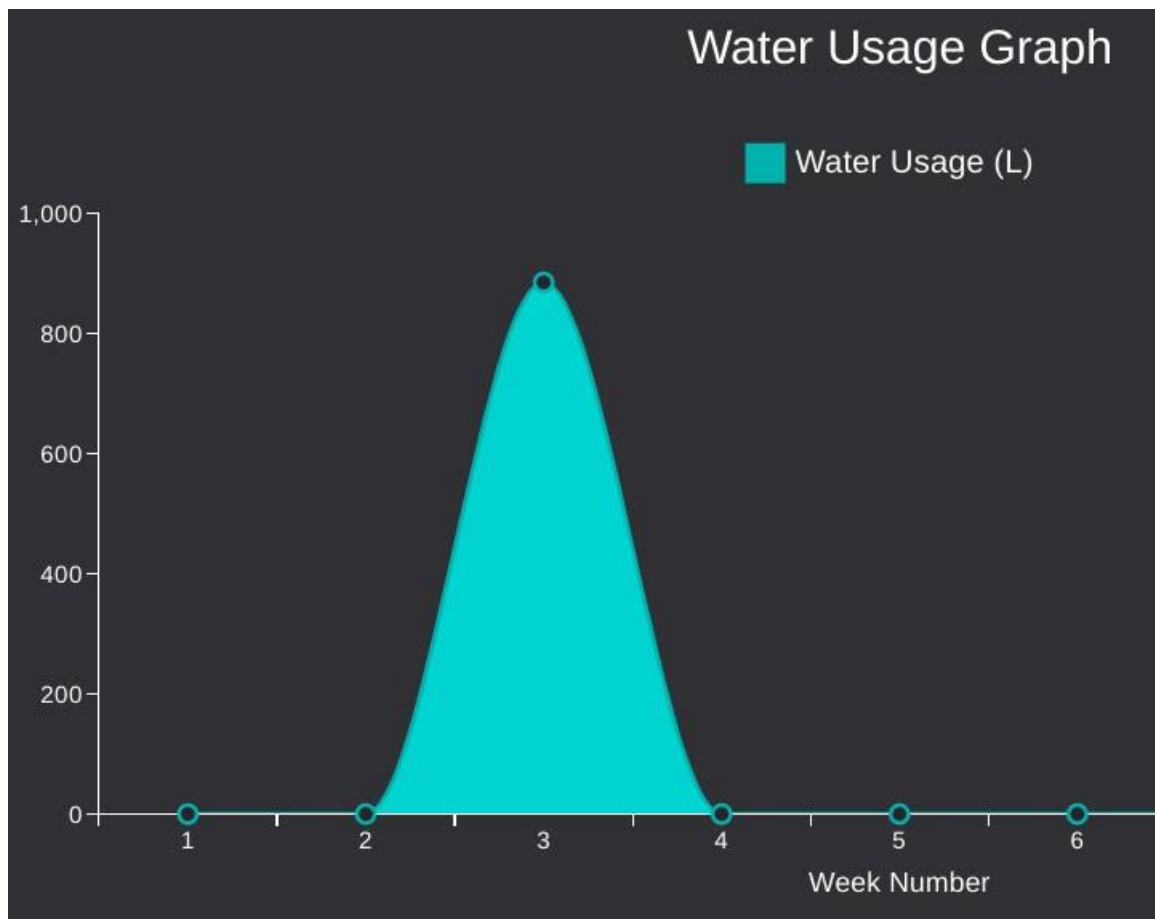


Figure 27: Water usage graph from week 2 to week 4

### Interoperability ( Data flow between each element of the system )

The entire Smart Irrigation System consists of three sub systems i.e. the IoT device or irrigation device deployed in the field, the nodejs backend server to control and monitor the irrigation device and also store and retrieve user data, this also provides the necessary data for the frontend web dashboard. Finally the python server which hosts the AI model for prediction. Now, all these systems need to communicate with each other in order to function properly.

To establish the seamless communication and data transfer between the systems, the REST api are used over http protocol. The IoT device monitors the local environment and sends the data to the nodejs server for processing, storage and predictions related tasks, it also handles the web dashboard. The nodejs server then sends the environmental data to the python server to use the AI model for making irrigation related prediction. Finally the nodejs server receives the decision from the AI model and then send the response back to the IoT device. Also the IoT device send the request to update the live status to the nodejs server, the server than keeps track of environmental statuses, irrigation schedules etc. And when the user navigates to the web based dashboard, the frontend system sends request to the same nodejs server to retrieve the stored data and statistics.

```

Codeium: Refactor | Explain | Generate Function Comment | ✕
int ApiClient::sendPostRequest(const std::string& endpoint, const std::map<std::string, std::string>& data, String& res) {
    HTTPClient http;
    std::string url = baseApi_ + "/intercom" + endpoint + "?identifier=" + identifier_;
    http.begin(url.c_str());
    http.addHeader("Content-Type", "application/json");

    JSONVar jsonDoc;
    for (const auto& pair : data) {
        jsonDoc[pair.first.c_str()] = pair.second.c_str();
    }

    String requestBody = JSON.stringify(jsonDoc);
    int httpCode = http.POST(requestBody);
    if (httpCode > 0)
        res = http.getString().c_str();
    else
        res = "Error on HTTP request";

    return httpCode;
}

```

Figure 28:IoT device: send http post request to the nodejs server

```

int ApiClient::predictIrrigation(const std::map<std::string, std::string>& data) {
    String response;
    int code = sendPostRequest("/system/irrigation", data, response);
    JSONVar jsonDoc = JSON.parse(response.c_str());

    if (code != 200)
        return code;

    if (JSON.typeof(jsonDoc) == "undefined" || !jsonDoc.hasOwnProperty("irrigate")) {
        return -2; // Error case
    }

    return int(jsonDoc["irrigate"]);
}

```

Figure 29:IoT device: send environmental data to nodejs for prediction

```

const callPredictionModel = async (cropName: string, cropDays: number, parameters: LiveStatusType) => {
  // call the AI model for irrigation prediction
  const req = await fetch(AI_SERVER_URL, {
    body: JSON.stringify({
      crop_type: cropName,
      crop_days: cropDays,
      ...parameters
    }),
    method: "POST"
  });

  const res = await req.json();
  return res.irrigate === 1;
};

```

Figure 30: Nodejs server: calling python server for AI prediction

```

get = (routerParam?: string | number, queryParams?: F) => {
  return client
    .get<R>(`${this.endpoint()}${routerParam ? "/" + routerParam : ""}`, {
      params: queryParams,
    })
    .then((res) => res.data);
};

post = (body?: S, option = defaultOption) => {
  return client
    .post<R>(this.endpoint(), body, option)
    .then((res) => res.data);
};

put = (routerParam: string | number, body?: S, option = defaultOption) => {
  return client
    .put<R>(`${this.endpoint()}/${routerParam}`, body, option)
    .then((res) => res.data);
};

```

Figure 31: Reactjs: apiClient class to handle http calls to nodejs REST api

```

Codeium: Refactor | Explain | Generate JSDoc | ✕
const useLiveStatus = (systemId: string) => {
  return useQuery({
    queryKey: [...LIVE_STATUS_CACHE_KEY, systemId],
    queryFn: () => liveStatusService.setSubroute(`/${systemId}`).get(),
    staleTime: 10 * 1000, // 10 seconds
  });
};

export default useLiveStatus;

```

Figure 32:fetching live status from nodejs server using react query and react hooks

This is how all the modules interoperate with each other and act as a single system i.e. Smart Irrigation System.

## 7.5 Challenges and Solutions

During development of the smart irrigation system, the following problems were encountered:

- **Library Compatibility Issues:** Majority of libraries for controlling hardware like RTC module were available for Arduino and was hard to use that in ESP32, as a solution server side time tracking was implemented.
- **Connectivity Issues:** Rural areas faced stable network connection issues, this was solved by implementing a fallback offline mode for the irrigation system using predefined threshold sensor values.
- **Power Management:** Improved power usage by limiting the total number of times the esp32 communicates with server.

Smart Irrigation System development managed to connect the IoT with AI and thus smartly and efficiently manage irrigation, as well as water resource. The iterative approach enabled the system to improve step by step over time and thus created a strong and reliable system.

## Chapter 8. Testing

Initially each sensors and components including temperature and humidity sensor, soil moisture sensor were connected to the microcontroller and then were tested to check if they work by writing a simple test programs i.e. just reading the sensor value and printing on the display.

```
void sensorTest(Sensors* sensors) {  
    auto moisture = sensors->readMoisture();  
    float temp, humidity;  
    bool m = sensors->readTempHumidity(&temp, &humidity);  
    if (!m) {  
        return;  
    }  
    Serial.println("Measured: " + String(m));  
    Serial.println("Moisture: " + String(moisture));  
    Serial.println("Temp: " + String(temp));  
    Serial.println("Humidity: " + String(humidity));  
    delay(500);  
}
```

Windsurf: Refactor | Explain | Generate Function Comment | ×

```
void motorTest(byte motorPin) {  
    Serial.println("Motor on");  
    digitalWrite(motorPin, HIGH);  
    delay(5000);  
    Serial.println("Motor off");  
    digitalWrite(motorPin, LOW);  
    delay(5000);  
}
```

Figure 33:Code Snippet to test the sensors reading and motor pump

```

Measured: 1
Moisture: 1023
Temp: 22.39
Humidity: 61.35
Measured: 1
Moisture: 1023
Temp: 22.42
Humidity: 61.33
Measured: 1
Moisture: 567
Temp: 22.44
Humidity: 61.36
Measured: 1
Moisture: 474
Temp: 22.41
Humidity: 61.31

```

Figure 34: Serial monitor console output from the esp32

To test the backend nodejs server which is the main controller component of the entire irrigation system, unit and integration testings were done using the vitest nodejs testing library.

Following are the major test cases that covers the most significant functionality of the system.

#### a. Authorization Middleware Test

Trying to access the currently installed IoT devices by calling the /api/systems route.

| S.N | Test Case                                                                                | Expected Result                                 | Actual Result |
|-----|------------------------------------------------------------------------------------------|-------------------------------------------------|---------------|
| 1   | Try to fetch the system info without logging in.                                         | It should return 401 i.e. unauthorized          | As expected   |
| 2   | Try to fetch the system info, after logging in, but by sending invalid session token.    | It should return 401 status code                | As expected   |
| 3   | Try to fetch the system info, after logging in, but this time with expired session token | It should return 401, unauthorized status       | As expected   |
| 4   | Try to fetch the system info, after logging in, and with valid session token             | It should return all the available systems info | As expected   |

Table 7: Authorization Middleware Test Case

## b. Login tests

Trying to test the login feature of the system if it properly works under any kind of input credentials. Try to login with different username and passwords and check if the system properly handles that.

| S.N | Test Case                                      | Expected Result                                                                     | Actual Result |
|-----|------------------------------------------------|-------------------------------------------------------------------------------------|---------------|
| 1   | Try to log in with correct credentials.        | It should return 200 status and include the username in the response.               | As expected   |
| 2   | Try to log in with incorrect credentials.      | It should return 401 status with an error message mentioning incorrect credentials. | As expected   |
| 3   | Check if sessionId is set on successful login. | The response should include a Set-Cookie header containing sessionId.               | As expected   |
|     |                                                |                                                                                     |               |

Table 8:Login tests case

## c. Crop Session Controller testings

This is the part where the user logs in to the system and then adds a new crop session. Crop session refers to the period of time where a specific crop is grown. User have to enter the details like which crop is s/he growing and the crop age if there is already a crop growing while installing the system, if the crop is to be cultivated later, the age count becomes 0. The system then monitors and irrigates the currently active session.

Following test cases cover if the system correctly handles creating a new crop session with various inputs.

| S.N | Test Case                                                  | Expected Result                                                            | Actual Result |
|-----|------------------------------------------------------------|----------------------------------------------------------------------------|---------------|
| 1   | Attempt to create a crop session without cropName.         | Should return 400 with a validation error mentioning cropName is required. | As expected   |
| 2   | Attempt to create a crop session without ageCount.         | Should return 400 with a validation error mentioning ageCount is expected. | As expected   |
| 3   | Attempt to create a crop session with an invalid cropName. | Should return 400 with a validation error mentioning Invalid crop name.    | As expected   |

|   |                                                            |                                                                                                                                     |             |
|---|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 4 | Attempt to create a crop session with a negative ageCount. | Should return 400 with a validation error mentioning ageCount cannot be negative.                                                   | As expected |
| 5 | Create a crop session with valid data.                     | Should return 200 and return the created session details (cropName, ageCount, cropSessionId, and systemId).                         | As expected |
| 6 | Verify the crop session is added to the database.          | The database should contain the created crop session with correct details (cropName, ageCount, initialCropAge, and lastAgeUpdated). | As expected |

Table 9: Crop Session Controller testings Case

#### d. Intercom Controller testings

This is the part of the server that actually communicates with the IoT device in the farmland. The IoT devices send multiple requests to this endpoints by sending the environmental conditions of the farmland. Since this is one of the most important part of the system i.e. receiving and updating the live status of the farmland so the user can view the real time status of the land.

Following tests cases ensure that the system correctly handles the communication with the remote IoT device and successfully updates the database and store the live status. The tests also ensures whether the system correctly authenticates the IoT device or not, and also if the system can handle incorrect inputs.

| S.N | Test Case                                                     | Expected Result                                                                    | Actual Result |
|-----|---------------------------------------------------------------|------------------------------------------------------------------------------------|---------------|
| 1   | Attempt to send live status without systemIdentifier.         | Should return 401 with an error mentioning system identification is required.      | As expected   |
| 2   | Attempt to send live status with an invalid systemIdentifier. | Should return 401 with an error mentioning system identification is invalid.       | As expected   |
| 3   | Attempt to send live status without irrigationStatus.         | Should return 400 with a validation error mentioning irrigationStatus is required. | As expected   |
| 4   | Attempt to send live status without temperature.              | Should return 400 with a validation error mentioning temperature is required.      | As expected   |
| 5   | Attempt to send live status without humidity.                 | Should return 400 with a validation error mentioning humidity is required.         | As expected   |

|   |                                                               |                                                                                                                                    |             |
|---|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 6 | Attempt to send live status without moisture.                 | Should return 400 with a validation error mentioning moisture is required.                                                         | As expected |
| 7 | Attempt to send live status with an invalid irrigationStatus. | Should return 400 with a validation error mentioning invalid irrigation status.                                                    | As expected |
| 8 | Send valid live status data.                                  | Should return 200 with a success message.                                                                                          | As expected |
| 9 | Verify live status update in the database.                    | The database should contain the updated live status with correct values for irrigationStatus, temperature, humidity, and moisture. | As expected |

Table 10: Intercom Controller testings Case

### e. System Controller testings

This is the part where the user registers the IoT device in the server which s/he just installed in the farmland. Since there can be multiple IoT devices even in the same field, they need to be registered into the server and also properly authenticated for their identification.

This testing ensures that the IoT device is properly registered into the server and also the server is able to handle inputs in different edge cases.

| S.N | Test Case                                               | Expected Result                                                                           | Actual Result |
|-----|---------------------------------------------------------|-------------------------------------------------------------------------------------------|---------------|
| 1   | Attempt to register a system without pumpFlowRate.      | Should return 400 with a validation error mentioning pumpFlowRate is required.            | As expected   |
| 2   | Attempt to register a system with systemName too short. | Should return 400 with a validation error mentioning name must be longer.                 | As expected   |
| 3   | Attempt to register a system with pumpFlowRate too low. | Should return 400 with a validation error mentioning it must be at least 0.1L/min.        | As expected   |
| 4   | Register a system with valid data.                      | Should return 200 with system details including systemName, pumpFlowRate, and identifier. | As expected   |
| 5   | Verify system registration in the database.             | Database should contain the new system with correct systemName and pumpFlowRate.          | As expected   |
| 6   | Verify related records are created upon registration.   | systems, liveStatus, systemSessions should contain relevant records.                      | As expected   |

|   |                                                  |                                                                  |             |
|---|--------------------------------------------------|------------------------------------------------------------------|-------------|
| 7 | Verify system session includes systemIdentifier. | systemSessions table should have an entry with systemIdentifier. | As expected |
|---|--------------------------------------------------|------------------------------------------------------------------|-------------|

Table 11: System Controller testings case

Above test cases summarizes almost all of the major functionalities of the irrigation system and make sure that the system is properly functioning without any errors. Below is the image of the final result of all those test cases code execution. For detailed screenshots of each of the test case code snippets, please refer to the appendix section.

```

✓ test/auth.test.ts (4)
✓ test/authController.test.ts (3) 440ms
✓ test/cropSessionController.test.ts (6) 1025ms
✓ test/intercomController.test.ts (9) 749ms
✓ test/systemsController.test.ts (5) 752ms

Test Files 5 passed (5)
  Tests 27 passed (27)
  Start at 20:11:50
  Duration 3.27s

PASS Waiting for file changes...
press h to show help, press q to quit

```

Figure 35: Result of test cases execution

## Chapter 9 : Evaluation

The evaluation phase of the Smart irrigation system includes its assessment based on system's usability, performance and reliability. The system was deployed for testing under real world conditions to ensure that all the elements like IoT devices, backend server and AI model met the operational expectations.

However due to geographical and time constraints the testing couldn't be done in real farm land. After consulting with my supervisor, a small indoor garden was set up inside the college premises itself and the irrigation system was deployed to monitor and irrigate the test garden. And for the user feedback and evaluation part, the system was exhibited in the science exhibition hosted by the Patan college on 31st March 2025 where many students including me presented our artifact to numerous users and stakeholders. This section of the report presents the user evaluations and the assessment of the system based on it's performance in the test environment.



Figure 36: System Deployment for testing and evaluation

### 9.1 User Evaluation

During the exhibition of the project, many users used and interacted with the system. Each one of them had their own opinion and feedback regarding the system. Some of those feedback are listed below.

| S.N | Full Name | Feedback / Opinion |
|-----|-----------|--------------------|
|     |           |                    |

|   |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Ram Kumar Acharya<br>(IT Student)          | <p>Appreciations: The system is responsive and for me it's easy to navigate. I liked the dashboard, it shows live status and water usage graph clearly. Logging in and registering a system was smooth.</p> <p>Criticisms: Some form validations error message weren't user-friendly. They just showed raw error messages, this can be improved by displaying user friendly error messages.</p>                                                                                                                                     |
| 2 | Harish Neupane<br>(Business Student)       | <p>Appreciations: I found the system simple and effective. The water tracking features are great to have, also its good to see AI and IoT integrating so well.</p>                                                                                                                                                                                                                                                                                                                                                                  |
| 3 | Kritika Thapa<br>(Agriculture Student)     | <p>Criticisms: Though the UI is feature wise good but lacks the modern looks and feels, and also it doesn't have a user friendly and intuitive design, it's confusing and overwhelming for farmers with limited knowledge. They need proper training before being able to use it.</p>                                                                                                                                                                                                                                               |
| 4 | Krishna Baral IT<br>(Farmer, no knowledge) | <p>Appreciations: It's really nice to see that agriculture being modernized, the hassle of manually irrigating field finally seems to be addressed. Irrigating the field as per specific plants needs is really a great idea.</p> <p>Criticisms: Only monitoring the environmental condition and deciding to irrigate is not enough. Irrigating just before rainfall maybe a waste of water. Also only controlling the irrigation aspect won't boost the crop productivity as it is heavily affected by flood and soil erosion.</p> |
| 5 | Parbati Paudel<br>(Farmer, Illiterate)     | <p>Appreciations: I am amazed to see machine automatically irrigating the field when necessary. This cuts off a lot of manual works.</p> <p>Criticisms: I cannot use this system on my own. Setup and configuration process is tough. Besides I cannot really use the dashboard too as there are just buttons with no visual feedback. It would be easy if there were images with manual, or voice support to help people like me.</p>                                                                                              |
| 6 | Niru Adhikari                              | <p>Appreciations: I loved the concept of irrigating field based on specific crop type and its age. Also it's feature to remotely monitor live status and also manually control the irrigation.</p>                                                                                                                                                                                                                                                                                                                                  |

|   |                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | (IT student with farming background)            | <p>It's a solid application for farmers with basic tech skills.</p> <p>Criticisms: The device and app itself is not yet usable for my parents in the village. It needs to integrate some visual user manuals, and more local language integration and voice support.</p>                                                                                                                                                                                                                                                                                                                                            |
| 7 | Komal Niraula<br>(Farmer, Limited IT knowledge) | <p>Appreciations: The feature to track the total water usage per crop harvest is really helpful along with the device automatically irrigating the field. As it tracks the water usage, it helps in pre-estimating the total water required for a crop cycle allowing us to more efficiently manage the water resources.</p> <p>Criticisms: Rather than having the complicated initial configuration or setup, it would be easy and helpful if the system had just one physical button to start it.</p>                                                                                                             |
| 8 | Hari Ghimire<br>(Agriculture Student)           | <p>Appreciations: I'm glad to see AI finally making it's way into agriculture and efficiently irrigating the field based on environmental conditions and crop types. This reduces a lot of manual tasks. Also the ability to remotely track the live status of the system and farmland is really appreciable.</p> <p>Criticisms: Right now the system only seems to control the water flow. System does not consider other factors like soil pH level, fertilizer requirements and weather data, which greatly hamper the crop production. Considering these factors would make the system truly smart farming.</p> |

*Table 12: User feedbacks and appreciations*

The above table only documents the feedback and opinion of some specific users, however the overall views and opinions of all the users are summarized below.

#### Appreciations:

- Users found the automated irrigation system easy and intuitive after initial training.
- User interface is considerable and is intuitive after initial manual or guidance.
- Some users with agricultural and science background claimed this would certainly reduce the water wastage, proving its effectiveness.
- The ability to remotely monitor the real time soil moisture, plant age and irrigation status was really appreciated.

- Majority of users liked the feature of remotely enabling/disabling the irrigation when needed bypassing the system decisions.
- Users found that the system would significantly reduce the manual work required to irrigate the field. And this was the most appreciated feature.

Criticisms:

- Almost all the users needed initial guidance or explanation to navigate the user interface, showing the need to further improve the UI to make it more intuitive and self-explanatory.
- Users with IT background criticized the manual override feature being remotely controlled, as this would generally fail in areas with limited internet connectivity.
- Users criticized the real time updated being delayed due to poor internet connection.
- Farmers with limited technical knowledge denied adopting it, as it required some technical knowledge for initial setup and configuration.
- Users without WIFI or internet connections found that the system has limited offline functionality and is barely usable.

## 9. 2 Performance Analysis

The performance testing mainly evaluates the system's responsiveness and how many request the system can handle per unit time. In this test, mainly two major aspect of the system are being tested. Since the server receives the request from users as well as from the IoT device, it is important to ensure that it can efficiently handle the request from both humans and irrigation devices.

The first case is the user logs in and views his/her list of installed systems, this ensures that the system can handle multiple users logging in to the system for remote monitoring.

The Second case is the IoT or irrigation device sending live status updates to the server. This ensures that many IoT devices can communicate to the server to update live status or make irrigation predictions.

```

config:
  target: "http://localhost:3000"
  plugins:
    expect: {}
  phases:
    - duration: 30
      arrivalRate: 20
    - duration: 30
      arrivalRate: 40
scenarios:
  - flow:
    - post:
      url: "/api/auth/login"
      json:
        username: "testuser"
        password: "testuser"
      capture:
        headers:
          Set-Cookie: "sessionId"
          as: "sessionId"
      expect:
        - statusCode: 200
    - get:
      url: "/api/systems"
      headers:
        Cookie: "sessionId={{sessionId}}"
      expect:
        - statusCode: 200
    - flow:
      - post:
        url: "/intercom/system/live?identifier=01960439-1b8d-788a-b637-0b9c8ed920f6"
        json:
          irrigationStatus: "OFF"
          temperature: 21
          humidity: 50
          moisture: 410
        expect:
          - statusCode: 200

```

Figure 37: Artillery load testing code snippets

The above figures is the code snippets to test the different routes of the server by sending multiple requests. The artillery framework is used to test the system performance.

Once the testing was executed the artillery generated a the test results into a JSON file, however the report was not user friendly and a bit overwhelming to include here. So, the AI tools like ChatGPT was used to generate a visual and user friendly information out of the test report and summarize is in a clear and concise manner.

Following is the performance test report of the system.

### System Load Test Summary

| Metric                      | Description                                     | Value           |
|-----------------------------|-------------------------------------------------|-----------------|
| Test Duration               | Total test running time                         | 60 seconds      |
| Total Requests Sent         | Sum of all HTTP requests sent during the test   | 30,000          |
| Total Successful Responses  | Number of 2xx (successful) responses            | 30,000          |
| Total Failed Responses      | Number of failed (non-2xx) responses            | 0               |
| Failure Rate (%)            | Percentage of failed responses                  | 0.00%           |
| Requests per Second         | Average throughput (requests/sec)               | 500             |
| Peak Concurrent Connections | Maximum number of simultaneous open connections | 14              |
| Reused Connections          | Connections reused instead of creating new ones | 99.91% (27 new) |
| Total Data Transferred      | Combined size of response bodies                | 7.35 MB         |
| Average Payload Size        | Average size of each response payload           | 255 bytes       |

Table 13: System Load Test Summary table

### Response Time Metrics (Latency)

| Metric                | Description                                    | Value    |
|-----------------------|------------------------------------------------|----------|
| Average Response Time | Overall mean response time across all requests | 9.54 ms  |
| Fastest Response Time | Minimum response time observed                 | 3.00 ms  |
| Slowest Response Time | Maximum response time observed                 | 65.00 ms |
| Standard Deviation    | Variance in response time                      | 4.32 ms  |
| P50 (Median)          | 50th percentile of response time               | ~9 ms    |
| P90                   | 90th percentile latency                        | ~14 ms   |
| P95                   | 95th percentile latency                        | 16.00 ms |
| P99                   |                                                | 25.00 ms |

Table 14: Response Time Metrics (Latency) Table

### HTTP Code Status Distribution

| Status Code | Description           | Count  | Percentage |
|-------------|-----------------------|--------|------------|
| 200         | Success               | 30,000 | 100%       |
| 4xx / 5xx   | Client / Server Error | 0      | 0%         |
| Total       |                       | 30,000 | 100%       |

Table 15: HTTP Code Status Distribution Table

After summarizing the above report, following things can be concluded.

The system's performance is excellent as average response time is under 10 milliseconds.

- The request failure rate is 0% which shows that system's robustness and its ability to operate efficiently even in high network traffic. Additionally, the system sustained a high load or traffic of about 500 requests per second without any latency spikes or poor performance.
- The latency metrics (P95: 16 ms, P99: 25 ms) also shows that the system highly consistent and stable.

The performance test indicates that the backend is stable, scalable, highly responsive with minimum failure rate, and production ready under tested conditions.

## Chapter 10 : Critical Analysis

Smart Irrigation System was developed using an iterative approach in different phases. The process included a thorough research and development, and field implementation. It started with the identification of inefficiencies in traditional irrigation, moving towards a solution integrating AI and IoT thereby automating the irrigation as per the specific needs of the crop.

Initial stages involved clearly defining the aims and objectives and project scopes, performing literature reviews, and selecting methodologies that aligns with precision irrigation and sustainable irrigation practices. And then gradually evolving the system into an intelligent, data-driven system that could interpret real-time environmental conditions and plant type & age then control the irrigation autonomously.

Hardware integration and establishing the seamless communication between ESP32, cloud server and AI model was the biggest challenge in development. Soil moisture, temperature, and humidity sensors required a proper calibration and map their reading to suitable threshold in order to feed into the AI model. Also maintaining a reliable connection between cloud server, AI model and IoT device was a challenging task. The ESP32 microcontroller and REST API were integrated to facilitate real-time data transmission and system responsiveness. Another significant aspect was the development of the AI model for predictive irrigation, requiring extensive data collection and refinement to achieve reliable decision-making.

As the system gradually took its form from basic prototype to operational artifact then employed in the real world for testing, some issues and areas for improvements were noticed. Issues with sensor inconsistencies, excessive network reliability, and scalability were addressed by iterative refinement. Likewise, fallback mechanisms for offline mode and power management are the some features that enhanced the system robustness.

Following are the analysis of the system based on it's strength, weaknesses and possible improvements:

Strengths:

- **AI-Powered Decision Making:** The system automatically adjusts irrigation schedules based on real-time sensor data, reducing water wastage.
- **IoT Integration:** Real-time data acquisition through sensors enables precise monitoring of soil and environmental conditions.

**Scalability:** The system can be scaled up to accommodate different kinds of crops, larger agricultural lands, and additional environmental parameters.

- **Automation:** Reduces manual labor through automating irrigation, benefiting farmers with limited access to labor.
- **Cost Efficiency:** Since it optimizes water utilization, the system reduces the operational cost for farmers in the long run.

#### Weaknesses:

- **Hardware Dependency:** System performance relies heavily on the precision and longevity of sensors that may require maintenance now and then.
- **Internet Connectivity Issues:** Since the system communicates with a cloud-based server, unstable internet connectivity can hinder its real-time decision-making capability.
- **Initial Setup Cost:** Although the system reduces long-term costs, the upfront cost of IoT devices and infrastructure can be expensive for small-scale farmers.

#### Possible Improvements:

- **Integration of Renewable Energy:** Incorporating solar energy for IoT devices can help increase sustainability and decrease the reliance on external power sources.
- **Machine Learning Enhancements:** Tuning AI models to fit different crop cycles, soil types, and weather patterns can continue to optimize irrigation.
- **Offline Functionality:** Building an offline mode wherein the system can operate based on cached data when the connectivity is lost.

## Chapter 11: Limitations and Future Scope

The project mainly focused on developing an affordable system targeting the resource constrained environment. On top of that the system was developed in the limited resources provided by the college in the given limited time frame. Due to which the system has many flaws and limitations that can be improved in the future enhancements. This section lists the project constraints and describes possible ways for it to be improved in the future.

### Limitations:

- **AI model with poor training data quality:** The model was trained in limited dataset and the accuracy of the AI model largely depends on the quality and quantity of data used in training, which might need continuous updating to avoid data skews.
- **Environmental Factors:** Unexpected climatic conditions such as heavy rainfall or severe drought might necessitate the installation of additional sensors in order to make the system more adaptive.
- **No user friendly setup:** Though the system once installed is autonomous, the initial setup and configuration can be overwhelming for non technical users.
- **Limited Crop Types:** The AI model currently only supports eight types of crop types, highly limiting it's applicability in diverse farming settings.
- **Limited field size:** Currently the system only performs well when used in small garden or farmland due to its limited sensors and peripherals.

### Future Scope:

- **Consider whether data:** Currently the system only considers local environmental data to decide irrigation needs which can be influenced heavily due to whether conditions like rainfall. In future the system can consider the rainfall prediction to avoid unnecessary irrigation.
- **App Integration that is Mobile-friendly:** Not everyone is comfortable with the web app. An easy to use mobile app can be developed for farmers allowing them to conveniently monitor the system status and also manually intervene the irrigation decisions if necessary.

**Support diverse crop types:** Increase the model dataset to support large number of crop types to increase the system's applicability in diverse agricultural settings.

- Support the large agricultural lands: Multiple sensors and water motor pumps can be installed across large fields to make irrigation decisions in each specific sections of the field, this ways the system can support large agricultural lands. The system should be scaled to add support for multiple sensors and pumps in future iterations.

## Chapter 12 : Conclusion

The Smart Irrigation System was developed to increase the efficient use of water and the health of crops by the help of AI and IoT technologies. This system operates the irrigation automatically according to the sensors data in the field, which constantly monitor soil moisture, temperature, humidity, and tracks the plant age. Thus, it optimizes the water use, thereby saving water and increasing crop yields, unlike conventional irrigation methods.

The entire process included AI models being trained for decision-making purposes and IoT devices being remotely monitored and controlled. The system provided high accuracy and a high level of responsiveness during its demonstration in a small environment in a short span of time. However its accuracy and robustness is yet to be tested in large environments over a long period of time.

### Key Findings:

- AI-driven automation significantly reduces water wastage.
- IoT integration allows remote monitoring and control, improving convenience.
- Scalability enables adaptation for different crops and climates.

### Contributions:

- Provides a sustainable irrigation model for smart agriculture.
- Introduces real-time analytics for decision-making.
- Reduces manual labor and increases efficiency.

### Outcomes:

- Demonstrated optimal water delivery to the crops thus saving water and increasing crop yields.
- Cost efficient
- Decreased manual human interventions in the field for irrigating.

Thus, this project can be viewed as a practical and Eco-friendly approach to agriculture which composes of various human technology interactions. It saves the humans a lot of manual work while only requiring sufficient resources. In the future, this can be further improved and enhanced leading towards a fully autonomous irrigation practices which rarely requires human interventions.

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## Chapter 14 : Appendix

### 14.1 Proposal Form RMET

#### BSc Project Screening Form: Guidelines

##### Part 1 – Project Proposal

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Student Name</b>            | Kushal Kafle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Student Number</b>          | 2133095                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Degree Pathway</b>          | Bsc (Hons) Computer Science & Software Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Supervisor Name</b>         | Krishna Aryal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Title of Project</b>        | Smart Irrigation System                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Abstract of the project</b> | <p>This research focuses on uses of IoT and AI in the development of smart systems that can also be implemented in the agricultural development to improve the irrigation system. The devices are to be assembled and integrated with the remote AI models, as per the requirement of this project research. As the traditional irrigation system faces many climatic challenges, and is also not very efficient when it comes to water resource management. This research tends to find an innovative approach to improve and optimize the water management in agriculture with the help of the project based on AI and IoT. The project includes the development of an irrigation system that is scalable and can be adapted to various agricultural settings. One of the goal of this research is finding the right sensors and AI models that can fit the requirements of project. It is also necessary to research on how the water usage analytics can be shown to the end user. This includes finding the right mathematical approach to calculate and estimate the water flow from the system and find out how much water is used by specific crop for the time being.</p> <p>The system interacts with various sensors to collect real time data on soil moisture, temperature and humidity. Then the collected sensor data is analysed by the AI model to make predictions on whether to start or stop the irrigation. Then system measures the total amount of water used for irrigation and produces analytics on the water used by various plants and in various years. Farmers can utilize this analytics and make informed decisions on which plants to grow in the future iterations and estimate the amount of water required for the plant. This ensures that the crops receive the optimal amount of water required for the proper growth, which leads to better crop productions.</p> <p><b>Keywords:</b> ESP32 based Irrigation; AI in irrigation; Smart Irrigation; Internet of Things in Agriculture; Predictive Irrigation System;</p> |

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project deliverables</b>         | <ul style="list-style-type: none"> <li>• Web Application for remote monitoring.</li> <li>• IoT based Demonstration</li> <li>• Final Report</li> <li>• Academic Poster</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Description of your artefact</b> | <p><b>Background:</b></p> <p>According to IOSR Journal (2015, pp. 32-36), With the water requirements in irrigation being large, there is a need for a smart irrigation system that can save about 80% of the water. Smart Irrigation can also prove to be efficient in Agricultural fields, Lawns &amp; Parks. It is achieved by installing sensors in the field to monitor the soil temperature and soil moisture which transmits the data to the microcontroller for estimation of water demands of plants.</p> <p>According to Choudhary et al. (2019), The autonomous irrigation system used in the study uses real time soil moisture data and AI based prediction algorithms to optimize water usage. The system analyses the historical weather data to predict the rainfall and climate change for more accurate water distribution. Tests conducted in controlled environment showed 80% accuracy which makes is effective solution for water conservation and improves the irrigation efficiency.</p> <p>According to Zhao et al. (2017), Water resource management is crucial in 21<sup>st</sup> century, as traditional agricultural irrigation systems are inefficient. IoT has introduced smart irrigation as a solution. This paper proposes a LoRa-based smart irrigation system, consisting of a LoRa communication module, solenoid valve, and hydroelectric generator. Data is transmitted wirelessly to the cloud through LoRa gateways, and the system can be controlled remotely via mobile applications. Experimental results demonstrate that the system is reliable in terms of both transmission distance and energy efficiency.</p> <p>According to Pernapati (2018), Since traditional irrigation methods consume a lot of water, there is a need of smart irrigation techniques to minimize the water waste. Smart irrigation system using IoT ans smart wireless sensors is implemented. Environmental conditions around the plants are monitored by the humidity, temperature and moisture sensors, and the ultrasonic sensor measures the water level in the reservoir. The system in controlled by ESP8266 NodeMCU microcontroller which processes the sensor data and then communicates via MQTT protocol to manage water usage efficiently.</p> <p>According to Sitharthan et al. (2023), An autonomous irrigation system is implemented using AI and 6G enabled IoT for smart agriculture. The prediction algorithm is used in the</p> |

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|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>microprocessor. The system uses historical weather data and IoT to predict rainfall and climate changes and then irrigates the crops accordingly. The system also considers soil moisture level to optimize the water supply. The system was tested with an accuracy of 86.34%, sensitivity of 89.28%, and precision of 91%.</p> <p><b>Problem Statement:</b></p> <p>This project addresses the challenges faced by existing traditional irrigation systems which are inefficient and do not consider the real time environmental conditions causing over irrigation leading to depletion of water resource or under irrigation leading to decreased crop productions. The project tries to solve the challenges faced by agricultural sectors due to global warming and climate change leading increased concerns about food security and water scarcity.</p> <p><b>Aim</b></p> <p>To research about the appropriate methods and approaches using IoT and AI in order to minimize the water wastage in agriculture and to increase agricultural productions to tackle global water scarcity and food security.</p> <p><b>Objectives</b></p> <ul style="list-style-type: none"> <li>→ Research and implement one of the approach or methods for smart irrigation using AI and IoT.</li> <li>→ Build a functional irrigation system using ESP32 to control water flow based on real-time sensor inputs.</li> <li>→ Implement an AI model to analyse the sensor data and predict the irrigation schedule.</li> <li>→ Measure the overall water usage in the given period by specific crops.</li> <li>→ Dashboard for monitoring the system as well as viewing the water usage analytics.</li> </ul> <p><b>List of features</b></p> <ul style="list-style-type: none"> <li>→ Automatic irrigation water flow based on AI predictions.</li> <li>→ Dashboard for remote monitoring of irrigation schedule and water usage analytics.</li> <li>→ Track the plant age in days and irrigate accordingly</li> <li>→ Option for manually entering the crop type and age.</li> </ul> |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                      | <p><b>Added value that the project provides</b></p> <ul style="list-style-type: none"><li>→ Supports further research in context of smart irrigation as it provides real world validation and testing.</li><li>→ Research points out the findings about the feasibility and challenges while implementing the smart irrigation in the real world.</li><li>→ The system tracks the water volume in each irrigation and gives detailed water usage analytics.</li><li>→ Remotely monitor the environmental conditions.</li><li>→ The system is cost effective as it uses components that are easily available at cheaper rate.</li><li>→ The system requires only one time installation or setup, and then it runs without any human intervention.</li></ul> <p><b>Intellectual challenges involved</b></p> <ul style="list-style-type: none"><li>→ <b>Sensor Integration and Calibration:</b> Integrating various sensors like soil moisture, temperature and humidity with ESP32 at different parts of the field and calibrating them for accurate data collection, and also handling the noise and errors in sensors in real time ensuring reliability at different environmental settings poses a technical challenge.</li><li>→ <b>Environmental Factors:</b> Since the system operates in open field, predicting the weather and rainfall is challenging due to their unpredictable nature, even weather forecasts are not reliably accurate. So the sudden environmental changes like rainstorms and fluctuating temperature may disrupt the irrigation schedule requiring AI model to quickly adapt and re-estimate the water needs dynamically which is a challenging task.</li></ul> |                   |                                                                                                                                            |                   |                            |                      |      |      |                                                                                                                                            |                    |      |      |                                                   |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------------------|----------------------|------|------|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------|------|------|---------------------------------------------------|
| <b>Risk Analysis</b> | <table><tr><th><b>Risk</b></th><th><b>Impact</b></th><th><b>Likelihood</b></th><th><b>Mitigation Strategy</b></th></tr><tr><td>Risk of Data Quality</td><td>High</td><td>High</td><td>It involves providing data validation checks, implement multiple sensors with options of data redundancy, and periodic sensor calibration.</td></tr><tr><td>Risk of Inaccurate</td><td>High</td><td>High</td><td>This can be mitigated by using multiple resources</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Risk</b>       | <b>Impact</b>                                                                                                                              | <b>Likelihood</b> | <b>Mitigation Strategy</b> | Risk of Data Quality | High | High | It involves providing data validation checks, implement multiple sensors with options of data redundancy, and periodic sensor calibration. | Risk of Inaccurate | High | High | This can be mitigated by using multiple resources |
| <b>Risk</b>          | <b>Impact</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Likelihood</b> | <b>Mitigation Strategy</b>                                                                                                                 |                   |                            |                      |      |      |                                                                                                                                            |                    |      |      |                                                   |
| Risk of Data Quality | High                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | High              | It involves providing data validation checks, implement multiple sensors with options of data redundancy, and periodic sensor calibration. |                   |                            |                      |      |      |                                                                                                                                            |                    |      |      |                                                   |
| Risk of Inaccurate   | High                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | High              | This can be mitigated by using multiple resources                                                                                          |                   |                            |                      |      |      |                                                                                                                                            |                    |      |      |                                                   |

|                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                               | <p>weather forecast</p> <p>for the weather forecast. Also, a hybrid approach with human and AI based prediction may be applied for the weather forecast.</p> <hr/> <p>Risk of Power Outage    Medium    High</p> <p>Implement backup power sources, use uninterruptible power supplies or solar powers.</p> <hr/> <p>Risk of sensor fault    High    Medium</p> <p>Implement sensor redundancy by utilizing different sensors for the validation of data, and perform regular maintenance along with calibration of the sensors.</p> <hr/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <p><b>How does your project relate to your degree course and build upon the units/knowledge you have studied/acquired</b></p> | <p>Throughout my Bachelor degree or course, I acquired a wide range of skill sets and knowledge in various fields. The Smart AI and IoT based irrigation system is built upon several areas of my studies. It integrates the concepts from AI, software engineering methodologies and programming.</p> <p><b>Principle of Programming</b></p> <p>My foundation on programming built by this course. It taught java programming in depth, which later helped me to further understand other programming languages and quickly learn them. It also helped me understand how programs communicate with the databases to store and retrieve the data.</p> <p><b>Fundamental Of Computer Studies</b></p> <p>In this course I learned how the computer hardwares work at a fundamental level. This course also taught low level programming at basic level which fascinated me even more to dive deep into microcontrollers programming. This is directly applicable to my current project while working with ESP32 microcontrollers and integrating sensors.</p> <p><b>OOP and Software Engineering</b></p> |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | <p>This course taught me software development lifecycle, design patterns and various data structures and algorithms. I also gained knowledge about cloud computing. Materials taught in this course are directly relevant to my project Smart Irrigation System.</p> <p><b>Research Methodologies and Emerging Technologies</b></p> <p>I learned about the systematic research and development aspect of the project in this course. I gained the knowledge to perform literature review, evaluate the emerging and innovative technologies and apply the appropriate and relevant research methods in the project I am developing. The Smart Irrigation System is built using the latest innovation in sustainable agricultural sector, AI and IoT.</p> <p>In conclusion, my academic courses has built my foundations in AI, computer hardwares, programming, software designing and engineering and software development. Each of these knowledge directly contributes to the development of Smart Irrigation System that is robust, adaptable, scalable and innovative solution for modern and sustainable agriculture.</p> |
| <b>Resources</b> | <p><b>Hardware Requirements</b></p> <ul style="list-style-type: none"> <li>→ ESP32 microcontroller</li> <li>→ Soil Moisture Sensor</li> <li>→ Temperature and humidity sensor</li> <li>→ Water Pump</li> <li>→ Relay Module</li> <li>→ Remote server with 4GB ram, and Intel core I5 or amd ryzen 5 for AI model</li> </ul> <p><b>Software Requirements</b></p> <ul style="list-style-type: none"> <li>→ Python IDE</li> <li>→ Python Interpreter 3.11</li> <li>→ AI and ML libraries (Scikit-learn, tensorflow, XGBoost, logistic regression, binary classification etc)</li> <li>→ Arduino IDE</li> <li>→ ESP32 drivers</li> </ul> <p><b>Languages Used</b></p> <ul style="list-style-type: none"> <li>→ C/CPP for ESP32 programming</li> <li>→ Python 3.11</li> <li>→ Postgresql (Database)</li> <li>→ ReactJS (for frontend interface)</li> </ul>                                                                                                                                                                                                                                                                           |

|                                                                                                                                                                              |                                      |                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|--------------------------|
| Have you completed & submitted your ethics form?                                                                                                                             | <input checked="" type="radio"/> YES | <input type="radio"/> NO |
| If the project is a development of previous work by yourself or others, give details below. Failing to declare such previous work here may be treated as an academic offence |                                      |                          |

Supervisor Signature:

*Kyala-*

29 october 2024

After the proposal has been signed off by both the supervisor and course coordinator scan the proposal and upload on BREO with signatures. Projects that follow proposals that have not been approved may be cancelled and there will be no compensation for any time lost

## **Part 2 – List of relevant resources**

### **1. Books**

- a. Kernighan, B.W. and Ritchie, D.M., 1988. *The C programming language*. 2nd ed. Englewood Cliffs, N.J.: Prentice Hall.
- b. Blum, R., 2014. *Arduino programming in 24 hours, Sams teach yourself*. Indianapolis: Sams Publishing.
- c. Barr, M., 1999. *Programming embedded systems in C and C++*. Sebastopol, CA: O'Reilly Media.

### **2. Journal Papers**

- a. IOSR Journal of Electronics and Communication Engineering (IOSR-JECE) e-ISSN: 2278-2834,p- ISSN: 2278-8735.Volume 10, Issue 3, Ver. II (May - Jun.2015), PP 32-36 [www.iosrjournals.org](http://www.iosrjournals.org); (Online)  
Available  
at :<https://smartfasal.in/wp/wp-content/uploads/2019/09/Smart-Irrigation-System.pdf>  
[Accessed on: 2024-10-02]
- b. Choudhary, S., Gaurav, V., Singh, A. and Agarwal, S., 2019. Autonomous crop irrigation system using artificial intelligence. *International Journal of Engineering and Advanced Technology (IJEAT)*, 8(5S), pp.46-50. (Online) Available at: <https://pdfs.semanticscholar.org/f4ec/2797766bd54505f7bbf9d4298d3765846ade.pdf>  
[Accessed on: 2024-10-04]
- c. Zhao, W., Lin, S., Han, J., Xu, R. and Hou, L., 2017. LoRa-based smart irrigation system. In: *2017 IEEE Globecom Workshops (GC Wkshps)*, 4-8 Dec. 2017, Singapore. Piscataway: IEEE. (Online) Available at: <https://ieeexplore.ieee.org/abstract/document/8269115>  
[Accessed on: 2024-10-04]
- d. Pernapati, K., 2018. Smart irrigation system using IoT and wireless sensors. In: *2018 Second International Conference on Inventive Communication and Computational Technologies (ICICCT)*, 20-21 April 2018, Coimbatore, India. Piscataway: IEEE. (Online) Available at: <https://library-guides.ucl.ac.uk/harvard/journal-article>

[Accessed on: 2024-10-04]

- e. Sitharthan, R., Rajesh, M., Vimal, S., Saravana Kumar, E., Yuvaraj, S., Abhishek Kumar, Raglend, J.I. and Vengatesan, K., 2023. A novel autonomous irrigation system for smart agriculture using AI and 6G enabled IoT network. *Microprocessors and Microsystems*, 101, p.104905. (Online) Available at:  
<https://www.sciencedirect.com/science/article/abs/pii/S0141933123001497>

[Accessed on: 2024-10-04]

- f. Agile Alliance (2023) Agile 101: What is Agile?. Available at:  
<https://www.agilealliance.org/agile101/> (Accessed: 25 October 2024)

## 14.2 Ethics Form

### FACULTY OF CREATIVE ARTS, TECHNOLOGIES AND SCIENCE

#### Form for Research Ethics Projects (Ethics Form)

|                      |                         |
|----------------------|-------------------------|
| Student Name         | Kushal Kafle            |
| Student Number       | 2133095                 |
| Degree Pathway       | Bsc (Hons) CS & SE      |
| Supervisor name      | Krishna Aryal           |
| Supervisor Signature | 29 October 2024         |
| Title of project     | Smart Irrigation System |

#### SECTION A Proposal

The Smart Irrigation System primarily involves experiments and testing in agricultural environments. It involves minimal human interaction and participation. However it may include some data collection from farmers or staffs for feedback, in this case, ethical considerations like informed consent, confidentiality and communication of research objectives shall be implied.

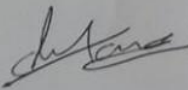
# SECTION B Check List

Please answer the following questions by circling YES or NO as appropriate.

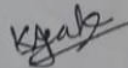
|                                                                                                                                                                |                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| Does the study involve vulnerable participants or those unable to give informed consent (e.g. children, people with learning disabilities, your own students)? | YES<br><input checked="" type="radio"/> NO |
| Will the study require permission of a gatekeeper for access to participants (e.g. schools, self-help groups, residential homes)?                              | YES<br><input checked="" type="radio"/> NO |
| Will it be necessary for participants to be involved without consent (e.g. covert observation in non-public places)?                                           | YES<br><input checked="" type="radio"/> NO |
| Will the study involve sensitive topics (e.g. obtaining information about sexual activity, substance abuse)?                                                   | YES<br><input checked="" type="radio"/> NO |
| Will blood, tissue samples or any other substances be taken from participants?                                                                                 | YES<br><input checked="" type="radio"/> NO |
| Will the research involve intrusive interventions (e.g. the administration of drugs, hypnosis, physical exercise)?                                             | YES<br><input checked="" type="radio"/> NO |
| Will financial or other inducements be offered to participants (except reasonable expenses or small tokens of appreciation)?                                   | YES<br><input checked="" type="radio"/> NO |
| Will the research investigate any aspect of illegal activity (e.g. drugs, crime, underage alcohol consumption or sexual activity)?                             | YES<br><input checked="" type="radio"/> NO |
| Will participants be stressed beyond what is considered normal for them?                                                                                       | YES<br><input checked="" type="radio"/> NO |
| Will the study involve participants from the NHS (patients or staff) or will data be obtained from NHS premises?                                               | YES<br><input checked="" type="radio"/> NO |

If the answer to any of the questions above is "Yes", or if there are any other significant ethical issues, then further ethical consideration is required. Please document carefully how these issues will be addressed.

Signed (student):  
Date: 2024-10-02



Countersigned (Supervisor):  
Date:



29 October 2024

### 14.3 Weekly Progress Status Forms from UG



**UNIVERSITY OF BEDFORDSHIRE**  
**DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY**  
**UG**  
**FINAL YEAR RMET PROJECT**  
**WEEKLY PROGRESS REPORT FORM**

|                                               |                                         |
|-----------------------------------------------|-----------------------------------------|
| Student's Name: <i>Kushal Katiel</i>          | Supervisor's Name: <i>Krishna Arsal</i> |
| Project Title: <i>Smart Irrigation System</i> |                                         |
| Week No: <i>1</i>                             | Report No: <i>1</i>                     |

|                                                 |                                                                                                                                                                            |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Summary of progress<br>(including any problems) | <i>→ Created frontend</i><br><i>→ Created backend</i><br><i>→ collected IoT devices &amp; sensors</i><br><i>→ collected data for model training</i>                        |
| Plan for next week                              | <i>→ test the sensor and developed integration.</i><br><i>→ start calibrating sensors</i><br><i>→ Train the AI model</i>                                                   |
| Supervisor's comments                           | <i>- collect data from sensor, (connect, get &amp; close)</i><br><i>- Pre-process &amp; send to db (RDBMS / NOSQL)</i><br><i>- start Documentation - Template (As per)</i> |

Student's Signature

Supervisor's Signature

*[Signature]*

*[Signature]* Day  
2025/1/16

Date *2025/01/16*

Date *Jan - 16, 2025*

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UNIVERSITY OF BEDFORDSHIRE  
DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY  
FINAL YEAR UG PROJECT  
WEEKLY PROGRESS REPORT FORM

|                                               |                                         |
|-----------------------------------------------|-----------------------------------------|
| Student's Name: <i>Kushal Kafle</i>           | Supervisor's Name: <i>Kritana Aryal</i> |
| Project Title: <i>Smart Irrigation system</i> |                                         |
| Week No: <i>02</i>                            | Report No: <i>02</i>                    |

|                                                 |                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Summary of progress<br>(including any problems) | <ul style="list-style-type: none"> <li>→ Create AI model using Random Forest.</li> <li>→ Train the model, achieved 98% accuracy.</li> <li>→ Create backend server for AI model deployment.</li> </ul>                                                                                                    |
| Plan for next week                              | <ul style="list-style-type: none"> <li>→ Integrate &amp; establish interoperability between node.js backend and Python AI model server.</li> <li>→ Test the functionality of IoT sensors.</li> </ul>                                                                                                     |
| Supervisor's comments                           | <ul style="list-style-type: none"> <li>→ Plan to execute Apply App Security</li> <li>→ <u>Documentation</u> <ul style="list-style-type: none"> <li>↳ Research on the final dissertation format/structure</li> <li>↳ Start the final documentation of completed Application System</li> </ul> </li> </ul> |

Student's Signature .....  
Date *2025/01/23*

Supervisor's Signature .....  
Date *24 JAN 2025*

*2025/1/26*

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DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY  
FINAL YEAR UG PROJECT  
WEEKLY PROGRESS REPORT FORM

|                                               |                                         |
|-----------------------------------------------|-----------------------------------------|
| Student's Name: <i>Kushal Kafle</i>           | Supervisor's Name: <i>Krishna Aryal</i> |
| Project Title: <i>Smart Irrigation System</i> |                                         |
| Week No: <i>03</i>                            | Report No: <i>03</i>                    |

|                                                 |                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Summary of progress<br>(including any problems) | <ul style="list-style-type: none"> <li>→ Started programming ESP32</li> <li>→ Scan, &amp; connect to wifi network</li> <li>→ Create storage to store preferences</li> <li>→ Create webserver for initial setup of IoT device.</li> </ul>                                                                                                                                        |
| Plan for next week                              | <ul style="list-style-type: none"> <li>→ create api client to send http requests to rest API.</li> <li>→ create toggle setup page switch.</li> <li>→ Handle connection errors with suitable indicators.</li> </ul>                                                                                                                                                              |
| Supervisor's comments                           | <ul style="list-style-type: none"> <li># Git/Dine share with Supervisor</li> <li># Environment Setup - Completed</li> <li># Collection - Gather of raw data, Process, Train</li> <li># 98% Accuracy</li> <li># Find document template (collect from faculty)</li> <li># Start the documentation of completed part</li> <li># Start UML (Class Diagram, ERM, Use case</li> </ul> |

Student's Signature ..... *Kushal* ..... Supervisor's Signature ..... *Krishna* .....  
Date *2025/02/30* ..... Date *2 Feb 2025* .....

*Ajay*  
*2025/2/3*

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FINAL YEAR UG PROJECT

WEEKLY PROGRESS REPORT FORM

|                                               |                                         |
|-----------------------------------------------|-----------------------------------------|
| Student's Name: <i>Isabel Kelle</i>           | Supervisor's Name: <i>Krithna Arjun</i> |
| Project Title: <i>Smart Navigation System</i> |                                         |
| Week No: <i>5</i>                             | Report No: <i>4</i>                     |

|                                                 |                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Summary of progress<br>(including any problems) | <ul style="list-style-type: none"> <li>→ Fix Arduino library incompatibility issue with ESP32.</li> <li>→ Test implement of kit with connection with error handling.</li> <li>→ Implement remote API call, establish server connection with error handling.</li> <li>→ Establish error connections.</li> </ul> |
| Plan for next week                              | <ul style="list-style-type: none"> <li>→ Complete the development part.</li> <li>→ Start setting of documentation.</li> </ul>                                                                                                                                                                                  |
| Supervisor's comments                           | <ul style="list-style-type: none"> <li>→ Add documentation in Git</li> <li>→ Work on Algorithm to increase the Accuracy to 99/100.</li> <li>→ Search the related dataset (Explore)</li> <li>→ Proceed to final documentations</li> </ul>                                                                       |

Student's Signature *[Signature]*  
Date *2025/02/12*

Supervisor's Signature *[Signature]*  
Date *Feb - 12-2025*

*[Signature]*  
*2025/2/13*

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DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY  
FINAL YEAR UG PROJECT  
WEEKLY PROGRESS REPORT FORM

|                                               |                                         |
|-----------------------------------------------|-----------------------------------------|
| Student's Name: <i>Kushal Kafle</i>           | Supervisor's Name: <i>Krishna Aryal</i> |
| Project Title: <i>Smart Irrigation System</i> |                                         |
| Week No: <i>6</i>                             | Report No: <i>5</i>                     |

|                                                 |                                                                                                                                                                                                                                                                |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Summary of progress<br>(including any problems) | <ul style="list-style-type: none"> <li>→ Complete frontend &amp; tested it.</li> <li>→ Complete backend &amp; tested it</li> <li>→ Complete IoT Programming.</li> <li>→ Integrate microcontroller with sensors, <del>test</del> and backend server.</li> </ul> |
| Plan for next week                              | <ul style="list-style-type: none"> <li>→ Start writing reflective report</li> <li>→ Start testing the device in indoor garden.</li> </ul>                                                                                                                      |
| Supervisor's comments                           | <ul style="list-style-type: none"> <li>- proceed for reflection report</li> <li>- " " final documentation as per template provided</li> <li>- code optimization.</li> </ul>                                                                                    |

Student's Signature ..... *[Signature]*  
Date *2025/02/19*

Supervisor's Signature ..... *[Signature]*  
Date *20 Feb 2025*

*copy*  
*2025/2/20*

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DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY

FINAL YEAR UG PROJECT  
WEEKLY PROGRESS REPORT FORM

|                                               |                                         |
|-----------------------------------------------|-----------------------------------------|
| Student's Name: <i>Kulshad Kaffle</i>         | Supervisor's Name: <i>Krishna Arora</i> |
| Project Title: <i>Smart Irrigation System</i> |                                         |
| Week No: <i>8</i>                             | Report No: <i>6</i>                     |

|                                                 |                                                                                                                                                                                                                                                               |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Summary of progress<br>(including any problems) | <ul style="list-style-type: none"> <li>→ Taken garden equipments for testing.</li> <li>→ Push existing code to git, get report samples.</li> <li>→ Started reflection report.</li> </ul>                                                                      |
| Plan for next week                              | <ul style="list-style-type: none"> <li>→ Complete <del>rest</del> of the reflection report.</li> <li>→ Deploy the dot in for testing in the garden.</li> </ul>                                                                                                |
| Supervisor's comments                           | <p><del>nothing</del></p> <ul style="list-style-type: none"> <li>- Finalize the final documentation as per UOB template</li> <li>- Finalize the application development</li> <li>- Poster/documentation by the end of week-9.</li> </ul> <p>Project (App)</p> |

Student's Signature *[Signature]*

Supervisor's Signature *[Signature]*

Date *2025/03/06*

Date *6 March 2025*

*[Signature]*  
*2025/3/6*

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DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY

<sup>UG</sup>  
FINAL YEAR ~~RMET~~ PROJECT

WEEKLY PROGRESS REPORT FORM

|                                               |                                         |
|-----------------------------------------------|-----------------------------------------|
| Student's Name: <i>Kushal Kafle</i>           | Supervisor's Name: <i>Krishna Aryal</i> |
| Project Title: <i>Smart Irrigation System</i> |                                         |
| Week No: <i>9</i>                             | Report No: <i>7</i>                     |

|                                                 |                                                                                                                                                                          |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Summary of progress<br>(including any problems) | <ul style="list-style-type: none"> <li>→ Complete Reflection Report</li> <li>→ Setup the project in a garden for testing.</li> </ul>                                     |
| Plan for next week                              | <ul style="list-style-type: none"> <li>→ Create the project Poster</li> <li>→ Start writing unit &amp; integration tests.</li> </ul>                                     |
| Supervisor's comments                           | <ul style="list-style-type: none"> <li>- finalize the app</li> <li>- finalize the Poster</li> <li>- proceed for final documentation &amp; presentation slides</li> </ul> |

Student's Signature ..... *[Signature]* .....

Supervisor's Signature ..... *[Signature]* .....

Date *2025/03/18*

Date *18 MARCH 2025*

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FINAL YEAR UG PROJECT  
WEEKLY PROGRESS REPORT FORM

|                                               |                                         |
|-----------------------------------------------|-----------------------------------------|
| Student's Name: <i>Kushal Kafle</i>           | Supervisor's Name: <i>Krishna Aryal</i> |
| Project Title: <i>Smart Irrigation System</i> |                                         |
| Week No: <i>10</i>                            | Report No: <i>8</i>                     |

|                                                 |                                                                                                                                                              |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Summary of progress<br>(including any problems) | → Create project poster<br>→ Write Unit/Integration tests of server using vitest framework.<br>→ Wrap up app                                                 |
| Plan for next week                              | → Start Final Thesis documentations                                                                                                                          |
| Supervisor's comments                           | - Project & Poster - Finalize by end of this week.<br>- Proceed to final documentation<br>- Demonstrate your final project<br>- Final presentation - proceed |

Student's Signature ..... *[Signature]*

Supervisor's Signature ..... *[Signature]*

Date ..... *2025/03/20*

Date ..... *20 MAR 2025*

When signed this form must be scanned and submitted via the relevant link on BREO

## 14.4 Google Forms Questionnaire

## Smart Irrigation System survey form

This is the survey questionnaire form to gather and collect information regarding user familiarity and interests regarding the smart technologies like precision irrigation in agriculture.

chrishdev.chd@gmail.com [Switch accounts](#)



 Not shared

Are you interested in making good income from you agricultural products ?

- ☐ Yes
- ☐ No
- ☐ Maybe

What is your primary role in agriculture ?

- ☐ Farmer
- ☐ Agriculture Consultant
- ☐ Researcher
- ☐ Government Official

Have you used any IoT-based devices in your farming operations ?

- ☐ Yes
- ☐ No
- ☐ Maybe

What factors would influence your decision to adopt a smart irrigation system ?

- ☐ Cost of the system
- ☐ Ease of use
- ☐ Reliability and accuracy
- ☐ Compatibility with existing infrastructure

How much are you willing to spend in devices like smart irrigation systems that automate your irrigation ?

- ☐ 2000 - 5000
- ☐ 5000 - 10000
- ☐ 10000 - 15000

What type of crops do you primarily grow ?

- ☐ Cereals (e.g. wheat, rice, millet)
- ☐ Fruits and vegetables
- ☐ Cash crops (e.g sugarcane, cotton)

What is the size of your agricultural land ?

- ☐ Less than 1 hectare
- ☐ 1 - 5 hectare
- ☐ 5 - 10 hectares
- ☐ More than 10 hectares

What irrigation method do you currently use ?

- ☐ Manual Irrigation
- ☐ Drip Irrigation
- ☐ Sprinkler Irrigation

How often do you monitor soil moisture levels before irrigating ?

- ☐ Never
- ☐ Occasionally
- ☐ Sometimes
- ☐ Often
- ☐ Always

What challenges do you face with your current irrigation system ?

- ☐ Over-Irrigation
- ☐ Under-Irrigation
- ☐ High water wastage
- ☐ High operating costs
- ☐ Lack of monitoring tools

Are you familiar with the concept of smart irrigation systems ?

- ☐ Yes
- ☐ No
- ☐ Maybe

How important is water conservation in your farming practices ?

- ☐ Not important
- ☐ Slightly important
- ☐ Moderately important
- ☐ Very important

Would you prefer an irrigation system that provides automated decision-making ?

- ☐ Yes, I prefer automation
- ☐ No, I prefer manual control
- ☐ Maybe, depending on the system's performance

What features would you like in a smart irrigation system ?

- ☐ Real time monitoring of soil, moisture and weather
- ☐ Automated irrigation based on environmental data
- ☐ Mobile app control and notifications
- ☐ Water usage statistics and reports
- ☐ Which crop is suitable for particular season

What challenges do you foresee in adopting smart irrigation systems ?

- ☐ Initial cost of setup
- ☐ Lack of technical knowledge
- ☐ Internet connectivity issues
- ☐ Maintenance requirements
- ☐ Illiterate individuals

Would you be willing to participate in a trial or demonstration of a smart irrigation system ?

- ☐ Yes
- ☐ No
- ☐ Maybe

## 14. 5 Models Accuracies and confusion matrix

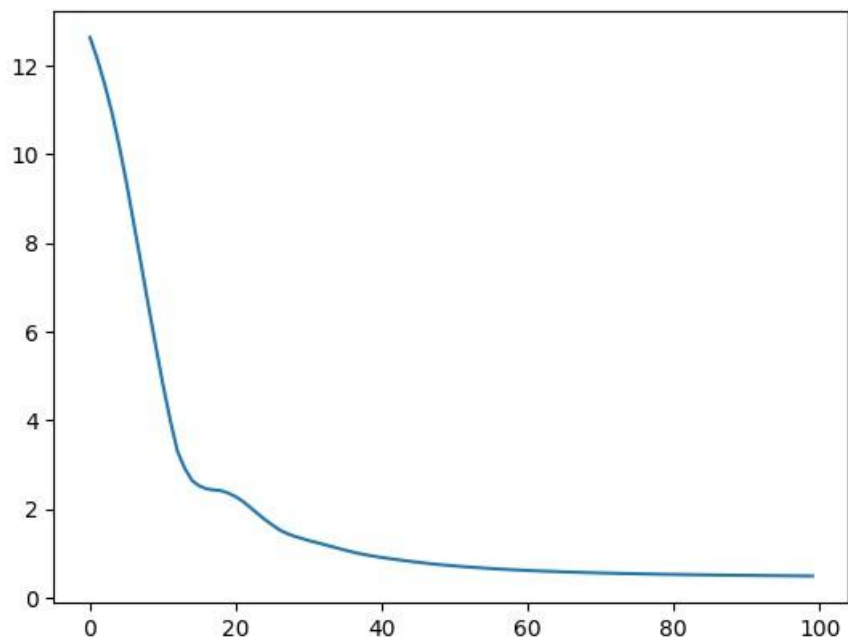


Figure 38: MLP Classifier: loss curve

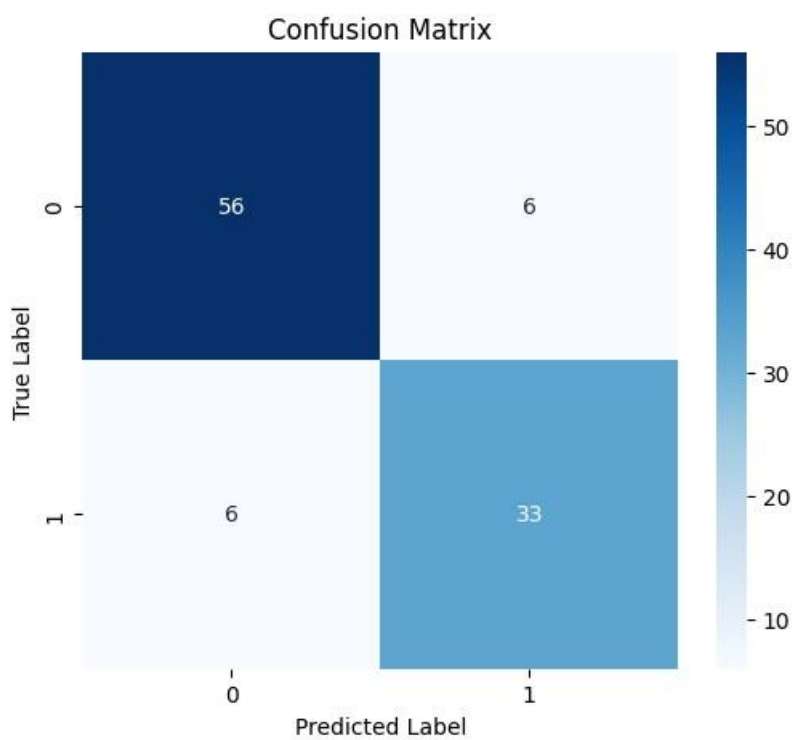


Figure 39: MLP Classifier: Confusion Matrix

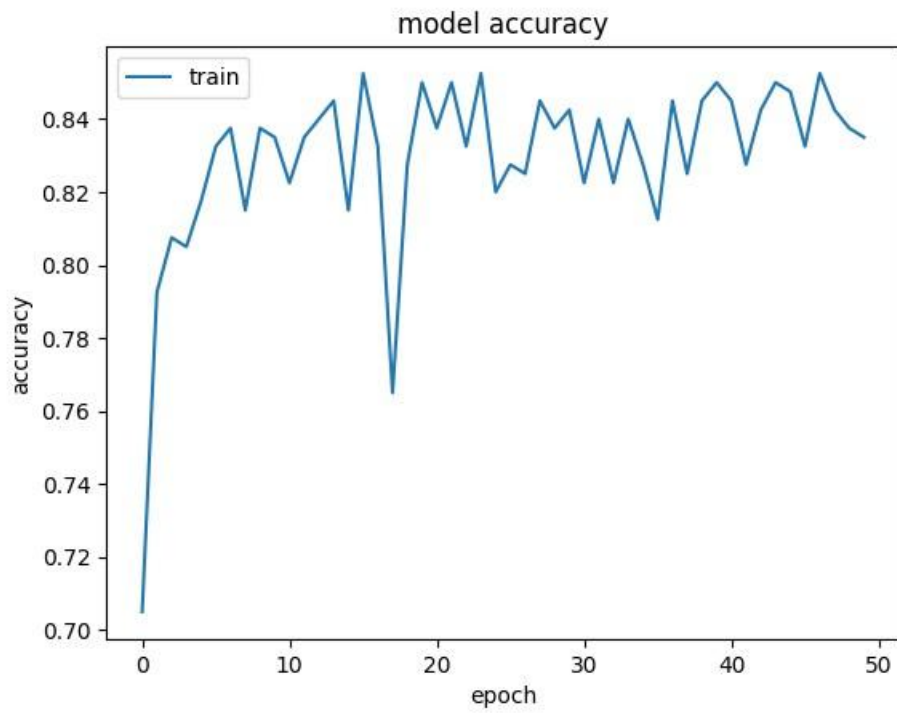


Figure 40: Keras: model accuracy

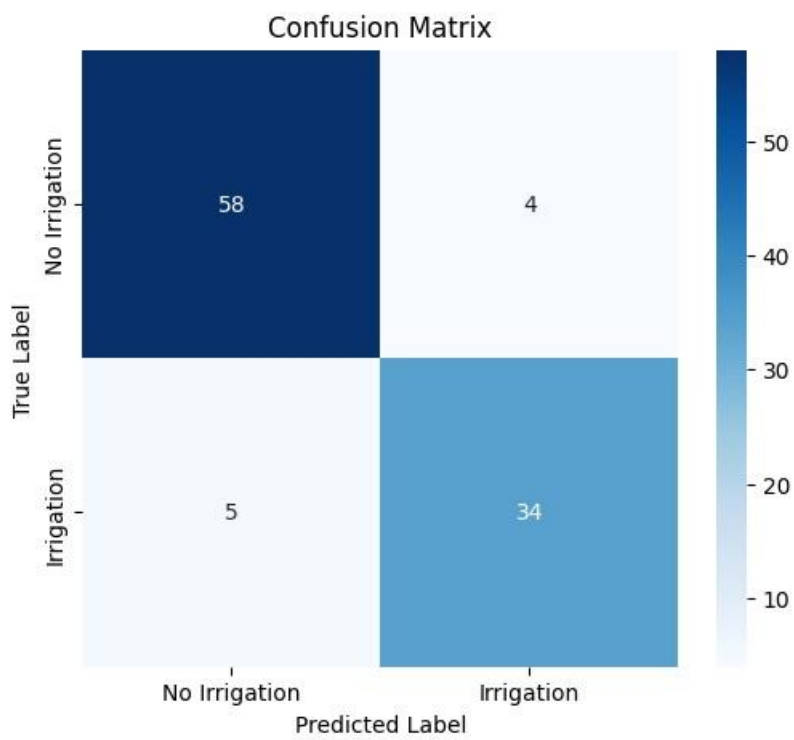


Figure 41: Logistic Regression: confusion matrix

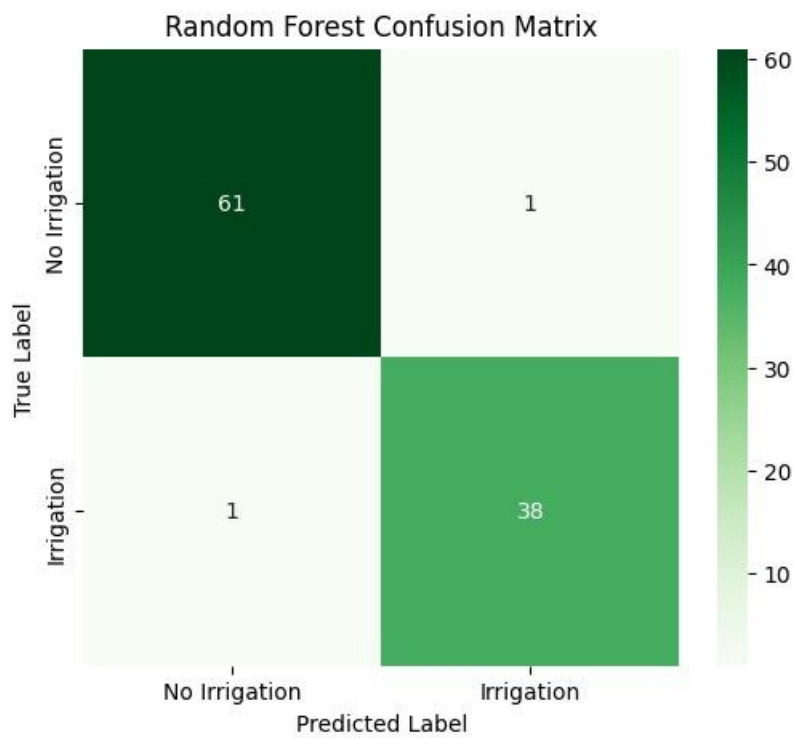


Figure 42:Random Forest: confusion matrix

## 14. 2 Test Cases code snippets

### 14.2.1 Authorization Middleware Test

```
it("should return 401 if not authorized", async () => {
  const res = await req.get("/api/systems");
  expect(res.status).toBe(401);
  expect(res.body.error).toBeTruthy();
});

it("should return 401 if invalid session id", async () => {
  const res = await req.get("/api/systems").set("Cookie", `sessionId=${uuidv4()}`);
  expect(res.status).toBe(401);
  expect(res.body.error).toBeTruthy();
});

it("should return systems info if valid session is given", async () => {
  const res = await req.get("/api/systems").set("Cookie", `sessionId=${sessionId}`);
  expect(res.status).toBe(200);
  expect(res.body).toHaveProperty("data");
});

it("should return 401 if expired session is given", async () => {
  // Make session expire
  const date = new Date();
  date.setDate(date.getDate() - 1);
  await prismaClient.sessions.updateMany({
    where: {
      session: sessionId,
    },
    data: {
      expiresAt: date,
    },
  });

  const res = await req.get("/api/systems").set("Cookie", `sessionId=${sessionId}`);

  expect(res.status).toBe(401);
  expect(res.body.error).toBeTruthy();
});
```

Figure 43: Code Snippets: Authorization middleware tests

### 14.2.2 Login Test

```
it('should login successfully with correct credentials', async () => {
  const response = await request(app)
    .post(loginUrl)
    .send({ username: testUser.username, password: testUser.password });

  expect(response.status).toBe(200);
  expect(response.body).toHaveProperty('username', testUser.username);
});

it('should fail login with incorrect credentials', async () => {
  const response = await request(app)
    .post(loginUrl)
    .send({ username: testUser.username, password: 'wrongPassword' });

  expect(response.status).toBe(401);
  expect(response.body).toHaveProperty('error');
  expect(response.body.error).toContain('Incorrect');
});

it('should set sessionId header on successful login', async () => {
  const response = await request(app)
    .post(loginUrl)
    .send({ username: testUser.username, password: testUser.password });

  expect(response.status).toBe(200);
  expect(response.headers['set-cookie']).toBeTruthy();
  expect(response.headers['set-cookie'][0]).toContain('sessionId');
});
});
```

Figure 44: Code Snippets: Login tests

### 14.2.3 Crop Session Controller Test

```
describe('POST /:systemId', () => {
  const getUrl = (id: string) => `/api/crop-sessions/${id}`;

  it('should throw validation error if cropName is missing', async () => {
    const response = await request(app)
      .post(getUrl(systemId))
      .set('Cookie', authCookie)
      .send({ ageCount: 5 }); // Missing cropName

    expect(response.status).toBe(400);
    expect(response.body).toHaveProperty('cropName');
    expect(response.body.cropName[0]).toContain('required');
  });

  it('should throw validation error if ageCount is missing', async () => {
    const response = await request(app)
      .post(getUrl(systemId))
      .set('Cookie', authCookie)
      .send({ cropName: CROP_TYPES[0] }); // Missing ageCount

    expect(response.status).toBe(400);
    expect(response.body).toHaveProperty('ageCount');
    expect(response.body.ageCount[0]).toContain('Expected');
  });

  it('should throw validation error if cropName is invalid', async () => {
    const response = await request(app)
      .post(getUrl(systemId))
      .set('Cookie', authCookie)
      .send({ cropName: 'InvalidCrop', ageCount: 5 }); // Invalid crop name

    expect(response.status).toBe(400);
    expect(response.body).toHaveProperty('cropName');
    expect(response.body.cropName[0]).toContain('Invalid');
  });
});
```

Figure 45: Code Snippets: Crop session controller test 1

```
it('should throw validation error if ageCount is negative', async () => {
  const response = await request(app)
    .post(getUrl(systemId))
    .set('Cookie', authCookie)
    .send({ cropName: CROP_TYPES[0], ageCount: -1 }); // Negative age count

  expect(response.status).toBe(400);
  expect(response.body).toHaveProperty('ageCount');
  expect(response.body.ageCount[0]).toContain('negative');
});

it('should create a crop session successfully with valid data', async () => {
  const response = await request(app)
    .post(getUrl(systemId))
    .set('Cookie', authCookie)
    .send(validCropSession);

  expect(response.status).toBe(200);
  expect(response.body[0]).toHaveProperty('cropName', validCropSession.cropName);
  expect(response.body[0]).toHaveProperty('ageCount', validCropSession.ageCount);
  expect(response.body[0]).toHaveProperty('cropSessionId');
  expect(response.body[0]).toHaveProperty('systemId', systemId);
});
```

Figure 46: Code Snippets: Crop Sessions Controller test 2

```

it('should add the crop session to the database', async () => {
  // Create a crop session
  await request(app)
    .post(getUrl(systemId))
    .set('Cookie', authCookie)
    .send(validCropSession);

  // Check if the crop session exists in the database
  const cropSessions = await prismaClient.cropSessions.findMany({
    where: {
      systemId,
      cropName: validCropSession.cropName,
    },
  });

  expect(cropSessions.length).toBe(1);
  expect(cropSessions[0]).toHaveProperty('cropName', validCropSession.cropName);
  expect(cropSessions[0]).toHaveProperty('ageCount', validCropSession.ageCount);
  expect(cropSessions[0]).toHaveProperty('initialCropAge', validCropSession.ageCount);
  expect(cropSessions[0]).toHaveProperty('lastAgeUpdated');
});
});
});

```

Figure 47: Code Snippets: Cropp Sessions Controller test 3

#### 14.2.4 Intercom Controller Test

```
describe('POST /system/live', () => {
  const url = '/intercom/system/live';

  it('should return 401 if system identifier is missing', async () => {
    const response = await request(app)
      .post(url)
      .send(validLiveStatus);

    expect(response.status).toBe(401);
    expect(response.body).toHaveProperty('error');
    expect(response.body.error).toContain('identify');
  });

  it('should return 401 if system identifier is invalid', async () => {
    const response = await request(app)
      .post(url)
      .query({ identifier: 'invalid-identifier' })
      .send(validLiveStatus);

    expect(response.status).toBe(401);
    expect(response.body).toHaveProperty('error');
    expect(response.body.error).toContain('identify');
  });

  it('should return validation error if irrigationStatus is missing', async () => {
    const response = await request(app)
      .post(url)
      .query({ identifier: testSystem.systemIdentifier })
      .send({
        temperature: validLiveStatus.temperature,
        humidity: validLiveStatus.humidity,
        moisture: validLiveStatus.moisture
      });

    expect(response.status).toBe(400);
    expect(response.body).toHaveProperty('status', 'error');
  });
});
```

Figure 48: Code Snippets: Intercom Controller test 1

```

it('should return validation error if humidity is missing', async () => {
  const response = await request(app)
    .post(url)
    .query({ identifier: testSystem.systemIdentifier })
    .send({
      irrigationStatus: validLiveStatus.irrigationStatus,
      temperature: validLiveStatus.temperature,
      moisture: validLiveStatus.moisture
    });

  expect(response.status).toBe(400);
  expect(response.body).toHaveProperty('status', 'error');
});

it('should return validation error if moisture is missing', async () => {
  const response = await request(app)
    .post(url)
    .query({ identifier: testSystem.systemIdentifier })
    .send({
      irrigationStatus: validLiveStatus.irrigationStatus,
      temperature: validLiveStatus.temperature,
      humidity: validLiveStatus.humidity
    });

  expect(response.status).toBe(400);
  expect(response.body).toHaveProperty('status', 'error');
});

it('should return validation error if irrigationStatus is invalid', async () => {
  const response = await request(app)
    .post(url)
    .query({ identifier: testSystem.systemIdentifier })
    .send({
      irrigationStatus: 'INVALID',
      temperature: validLiveStatus.temperature,
      humidity: validLiveStatus.humidity,
      moisture: validLiveStatus.moisture
    });
});

```

Figure 49: Code Snippets: Intercom Controller test 2

```

    expect(response.status).toBe(400);
    expect(response.body).toHaveProperty('status', 'error');
  });

  it('should return success response with valid data', async () => {
    const response = await request(app)
      .post(url)
      .query({ identifier: testSystem.systemIdentifier })
      .send(validLiveStatus);

    expect(response.status).toBe(200);
    expect(response.body).toHaveProperty('status', 'success');
  });

  it('should update the database with new live status', async () => {
    // Send the request with valid data
    await request(app)
      .post(url)
      .query({ identifier: testSystem.systemIdentifier })
      .send(validLiveStatus);

    // Check if the live status was updated in the database
    const updatedLiveStatus = await prismaClient.liveStatus.findUnique({
      where: {
        systemId
      }
    });

    expect(updatedLiveStatus).not.toBeNull();
    expect(updatedLiveStatus?.irrigationStatus).toBe(validLiveStatus.irrigationStatus);
    expect(updatedLiveStatus?.temperature).toBe(validLiveStatus.temperature);
    expect(updatedLiveStatus?.humidity).toBe(validLiveStatus.humidity);
    expect(updatedLiveStatus?.moisture).toBe(validLiveStatus.moisture);
  });
});

```

Figure 50: Code Snippets: Intercom Controller test 3

### 14.2.5 System Controller Test

```
describe('POST /register', () => {
  it('should throw validation error if not all required fields are given', async () => {
    const response = await request(app)
      .post(registerUrl)
      .set('Cookie', authCookie)
      .send({ systemName: 'Test System' });

    expect(response.status).toBe(400);
    expect(response.body).toHaveProperty('pumpFlowRate');
    expect(response.body.pumpFlowRate[0]).toContain('Expected');
  });

  it('should throw validation error if systemName is too short', async () => {
    const response = await request(app)
      .post(registerUrl)
      .set('Cookie', authCookie)
      .send({ systemName: 'Te', pumpFlowRate: 10.5 });

    expect(response.status).toBe(400);
    expect(response.body).toHaveProperty('systemName');
    expect(response.body.systemName[0]).toContain('characters');
  });

  it('should throw validation error if pumpFlowRate is too low', async () => {
    const response = await request(app)
      .post(registerUrl)
      .set('Cookie', authCookie)
      .send({ systemName: 'Test System', pumpFlowRate: 0.05 });

    expect(response.status).toBe(400);
    expect(response.body).toHaveProperty('pumpFlowRate');
    expect(response.body.pumpFlowRate[0]).toContain('0.1L/min');
  });

  it('should register the system if all fields are given correctly', async () => {
    const response = await request(app)
      .post(registerUrl)
      .set('Cookie', authCookie)
      .send(validSystem);
  });
});
```

Figure 51: Code Snippets: Systems Controller test 1

```

expect(response.status).toBe(200);
expect(response.body[0]).toHaveProperty('systemName', validSystem.systemName);
expect(response.body[0]).toHaveProperty('pumpFlowRate', validSystem.pumpFlowRate);
expect(response.body[0]).toHaveProperty('identifier');
});

it('should save the system in database once registration is successful', async () => {
  // Register a system
  await request(app)
    .post(registerUrl)
    .set('Cookie', authCookie)
    .send(validSystem);

  // Check if the system exists in the database
  const systems = await prismaClient.systems.findMany({
    where: {
      systemName: validSystem.systemName,
    },
  });

  expect(systems.length).toBe(1);
  expect(systems[0]).toHaveProperty('systemName', validSystem.systemName);
  expect(systems[0]).toHaveProperty('pumpFlowRate', validSystem.pumpFlowRate);

  // Check if related records were created
  const preferences = await prismaClient.systems.findFirst({
    where: {
      systemId: systems[0].systemId,
    },
  });
  expect(preferences).toBeTruthy();

  const liveStatus = await prismaClient.liveStatus.findFirst({
    where: {
      systemId: systems[0].systemId,
    },
  });
});

```

Figure 52: Code Snippets: Systems Controller test 2

```

expect(liveStatus).toBeTruthy();

const session = await prismaClient.systemSessions.findFirst({
  where: {
    systemId: systems[0].systemId,
  },
});
expect(session).toBeTruthy();
expect(session).toHaveProperty('systemIdentifier');
});
});
});

```

Figure 53: Code Snippets: Systems Controller test 3

## 14.3 User Interface

Smart Irrigation

Select a crop type

Age Count

Create

Search Crop Sessions

Sugarcane

Potato

Pulse

Groundnuts

Add Crop Session

Pulse

Age Count

100

Edit

Update

Deactivate

Delete

Smart Irrigation

Pulse Session Statistics

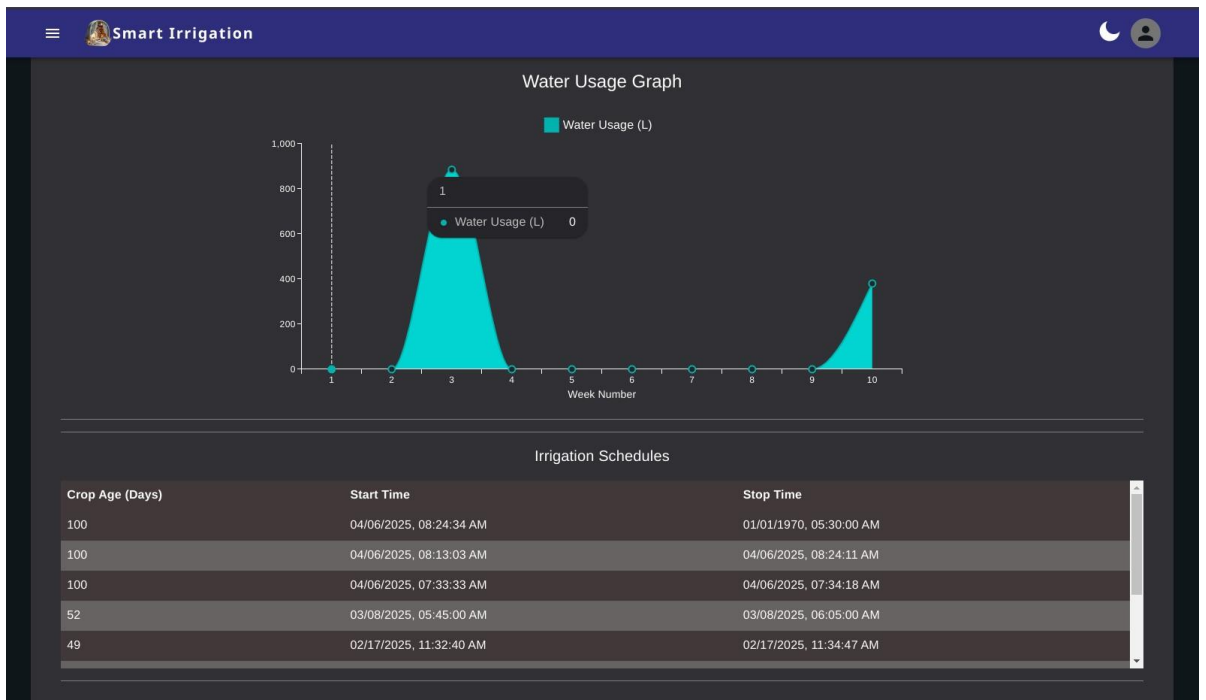
|                      |                  |
|----------------------|------------------|
| Operation Days       | 69 Days          |
| Operation Weeks      | 10 Weeks         |
| Irrigation Duration  | 0.56682 Hours    |
| Total Water Consumed | 1088.2944 Litres |

Irrigation Schedules

| Crop Age (Days) | Start Time              | Stop Time               |
|-----------------|-------------------------|-------------------------|
| 100             | 04/06/2025, 08:24:34 AM | 01/01/1970, 05:30:00 AM |
| 100             | 04/06/2025, 08:13:03 AM | 04/06/2025, 08:24:11 AM |
| 100             | 04/06/2025, 07:33:33 AM | 04/06/2025, 07:34:18 AM |
| 52              | 03/08/2025, 05:45:00 AM | 03/08/2025, 06:05:00 AM |
| 49              | 02/17/2025, 11:32:40 AM | 02/17/2025, 11:34:47 AM |
| 49              | 02/17/2025, 08:38:55 AM | 02/17/2025, 08:39:59 AM |
| 49              | 02/17/2025, 08:29:05 AM | 02/17/2025, 08:29:30 AM |
| 49              | 02/17/2025, 08:24:06 AM | 02/17/2025, 08:27:24 AM |
| 49              | 02/17/2025, 08:01:41 AM | 02/17/2025, 08:02:26 AM |

Visualize Statistics

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Smart Irrigation

### Farmland System

|                   |                         |
|-------------------|-------------------------|
| Crop              | Pulse                   |
| Age               | 100 Days                |
| Temperature       | 22 °C                   |
| Humidity          | 27 %                    |
| Moisture          | 837                     |
| Irrigation Status | ON                      |
| Last Updated      | 04/06/2025, 08:26:35 AM |

### Irrigation Schedules

| Crop Age (Days) | Start Time              | Stop Time               |
|-----------------|-------------------------|-------------------------|
| 100             | 04/06/2025, 08:24:34 AM | 01/01/1970, 05:30:00 AM |
| 100             | 04/06/2025, 08:13:03 AM | 04/06/2025, 08:24:11 AM |
| 100             | 04/06/2025, 07:33:33 AM | 04/06/2025, 07:34:18 AM |
| 52              | 03/08/2025, 05:45:00 AM | 03/08/2025, 06:05:00 AM |
| 49              | 02/17/2025, 11:32:40 AM | 02/17/2025, 11:34:47 AM |

 Smart Irrigation

Add new System

System Name

Demo System

Pump Flow Rate

3

0196140c-e393-777c-b982-7de6b0516926 [Copy](#)

[Back](#) [Submit](#)

 Smart Irrigation

Full Name

Username

Password

Confirm Password

[Register](#)

Smart Irrigation

System Name

Farmland System

Pump Flow Rate

32

Identifier:

0194f98e-7018-7554-82ce-f133f9dc72ad

Edit

Submit

Delete

Manage Sessions

Live Status

View Statistics

Preferences

Smart Irrigation

System Statistics

Irrigation Duration:

0.6588830555555555 Hours

Operation Days:

81

Operation Weeks:

12

Total Water Usage:

1265.0554666666665 Litres

Crop

Age (Days)

Start Date

Sugarcane

1

2/20/2025

Potato

1

12/30/2024

Pulse

100

12/30/2024

Groundnuts

6

12/30/2024

Smart Irrigation

Search Systems

Search

device

Farmland System

final system

hey system

test system 2

test system

Add System